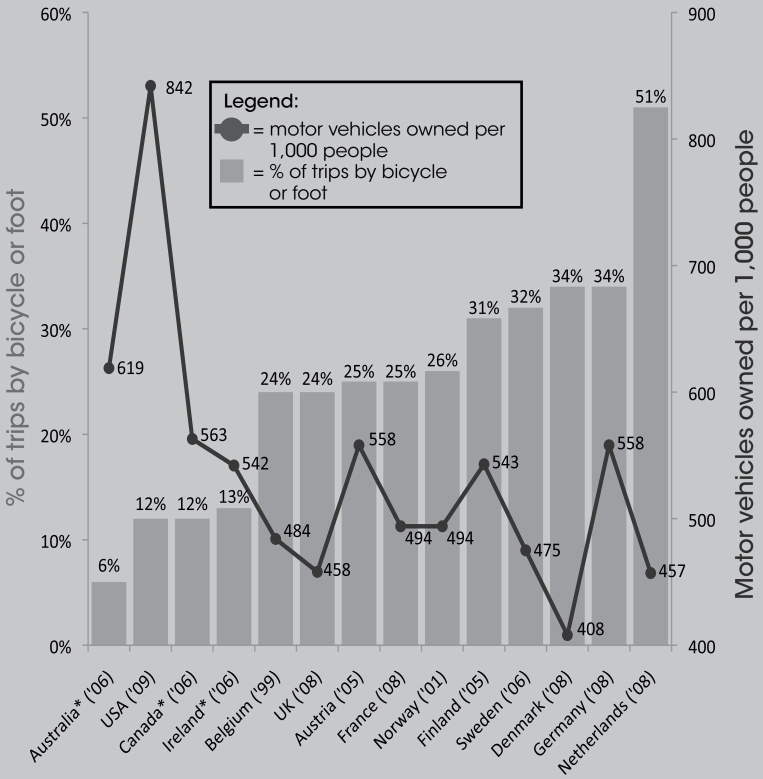
Looking Outside the Borders

Car Ownership and Bicycling and Walking Levels in Developed Nations

International data further suggest a correlation between low car ownership and higher levels of bicycling and walking.



Sources: J. Pucher and R. Buehler, 2010. "Walking and Cycling for Healthy Cities," *Built Environment 36*(5), pp. 391-414, http://policy.rutgers.edu/faculty/pucher/BuiltEnvironment_WalkBike_10Dec2010.pdf and NationMaster, http://www.nationmaster.com/graph/tra_mot_veh-transportation-motor-vehicles#source. **Note:** *r* = -0.59.

collected for the 2010 report and this current report, there is still a severe deficiency in evaluation of these efforts. Because many cities and states could not provide data on participation levels, and many programs are brand new, it is difficult to explore potential relationships. The Benchmarking Project will continue to collect data on education and encouragement efforts and hopes to explore the relationship further in future benchmarking reports.

Looking to the Leaders

Case studies and closer looks at cities, states, and countries that have the highest bicycling and walking levels may help reveal what factors are most important. This report looks at two leaders—Oregon and Seville, Spain—and explores what factors might influence their growing levels of bicycling and walking.

Looking Outside the Borders

The bicycle boom in Seville, Spain

by Zach Vanderkooy, Bikes Belong Foundation

Seville's embrace of the bicycle is decidedly 21st century. As recently as 2004, bicycling in this city of 700,000 was seen as a fringe activity for elite athletes and people too poor to own a car. There was no bicycle infrastructure to speak of, and the few Sevillianos who did use bikes for utilitarian purposes (0.2% of all trips in 2000) were practically invisible on the streets and in public life. Cars and trucks dominated the transportation landscape. For the average person to ride a bike to work or school was unimaginable.jgj

For people living in most American cities, this story feels awfully familiar.

That's why Seville's remarkable transformation is drawing excited attention on our side of the Atlantic. In just five years, bicycling has grown from a statistically nonexistent mode of transportation to a significant—if not yet ordinary—part of daily life. Seville's engineers built a network of comfortable separated bikeways connecting the city that now carries 7% of all traffic. It has implemented a state-of-the-art bike sharing system, offering residents and visitors affordable access to more than 2,000 bicycles stationed throughout the city. And it has redesigned many plazas, squares, and streets to make them more inviting spaces for those traveling on foot and on two wheels.

The investments are paying dividends more quickly than anyone figured.

Bike lane in Seville, Spain. Photo by UncleBucko @ Flickr.

Traffic congestion and pollution are declining for the first time in 30 years. Businesses are thriving along bike routes and around the newly improved public spaces that are breathing fresh life into the central city. The number of car trips into the historic city center has plummeted from 25,000 a day to 10,000, freeing valuable space for residents to park and visitors to linger. More than 70,000 bike trips are made every day, up from just 2,500 in 2002. Bicycling has given Sevillianos a healthy, speedy new way to get around.

Designing a better city

When it came to designing Seville's bicycle infrastructure, the city looked north for inspiration. David Muñoz de la Torre, Director of Seville's Bike Program, cites the Netherlands as a key influence in shaping the city's bikeway system. "We wanted to create a complete network, not a piecemeal system," he emphasized. Nothing discourages potential bike riders more than bike lanes that end abruptly, forcing them to mix with fast-moving cars. Seville's bikeways continue through intersections, around roundabouts, and across zigzags, making navigation a breeze — just follow the impossible-to-miss bright green path and you'll stay on route.

All the bikeways along major roads are physically separated from car and pedestrian traffic as much as possible. City leaders explain that protection from traffic is a key factor in making the system appealing to less experienced riders, particularly children, women, and older people. "Our design target is a 65-year old woman with groceries," explained Muñoz de la Torre. He reasons that if bicycling is safe for her, it is safe for everyone.

Quick Facts

- Seville added 87 miles of new bike infrastructure in 36 months between 2007 and 2009.
- 85% of the space came from removal of car parking and travel lanes; 15% came from pedestrian space (which was compensated for by major, new additions to public space in other places).
- The improvements increased the percentage of all trips taken by bicycle from 0.4% to ~ 7%.
- In 2005, the Alameda de Hercules, a major public plaza, was redesigned to be more inviting to people on bikes and on foot. More than 100 public meetings were held to discuss the plan, which required reallocating 200 parking spaces to make room for new public space. Initially neighbors and local businesses strongly opposed the changes, but now 22% of customers arrive by bike and businesses along the plaza are thriving.
- City Council passed a law that restricts non-resident auto access into the cramped central city; the law reduced the daily number of cars in downtown from 25,000 to 10,000, drastically reducing congestion.
- "Great is the enemy of good." The city's infrastructure emphasizes network connectivity, not perfection. It's far from the polished bikeways of Northern Europe, but the protected bikeways of Seville are safe, convenient, and get you where you need to go without interruption.

The entire 87-mile network cost about $43 million to install between 2007 and 2009—a bargain when you consider that a single mile of urban freeway in the United States easily costs twice as much. With the core bicycling network now in place, Muñoz de la Torre says the city is focused on improving difficult crossings and tight squeezes on the streets, installing more bike parking, and launching public education campaigns as the next steps to boost bicycle use. The city expects 15% of all trips in Seville to be made by bike in 2015.

Change isn't easy—but it can happen overnight

In order to build support for its rapid urban transformation, Seville's leaders had to win the favor of a public that had little familiarity with bicycling and merchants skeptical that customers would arrive on bike and foot instead of in cars. The city hosted hundreds of public meetings and charrettes—design workshops involving the public—to incorporate ideas from neighborhoods into plans for newly configured public spaces and roadways.

Initial opposition to removing or relocating car parking was fierce, but business owners came to realize that streets filled with pedestrians and bicyclists create more opportunities for folks to spontaneously stop in a shop or café. Some controversy remains, but polls show both residents and businesses are predominantly pleased with the changes. The rise in bicycling is a bright spot in tough economic times, as stores, restaurants, and plazas of the central city are usually packed with residents and tourists.

If they can do it, why can't we?

For U.S. cities just beginning to build bikeway networks, Seville is an inspiring example of how quickly results can be achieved with focused investments. This scorching hot, car-centered Spanish city is far different from the Dutch and Danish cities usually celebrated as bicycling Meccas. Seville's story challenges the common assumption that biking and walking have always been a way of life in European cities. With families strolling and bikes rolling on avenues that just 5 years ago were filled with roaring cars and trucks, it's impossible not to ask the question: If Seville can do it, why not Dallas, Atlanta, or Los Angeles?

A row of Servici bicycles in Seville, Spain.
Photo Courtesy of lasgalletas @ Flickr.

Photo by donjd2 @ Flickr

8: PUBLIC HEALTH BENEFITS

Walking and bicycling have great potential to improve public health (Buehler et al., 2011, Gordon-Larsen 2009, Hamer and Chida 2008, Oja 2011, Pucher et al., 2010, Shephard 2008). Further, the health benefits of active transportation can outweigh any risks associated with these activities by as much as 77 to 1 (Rojas-Rueda et al., 2011).

In 2009, 40% of trips in the United States were shorter than 2 miles and 27% were shorter than 1 mile. Since bicycling can accommodate trips of up to 2 miles and most people can walk at least 1 mile, there is a lot of hope to use this form of travel in our communities. Still, Americans use their cars for 62% of trips up to 1 mile long and 87% of trips 1 to 2 miles long (NHTS 2009).

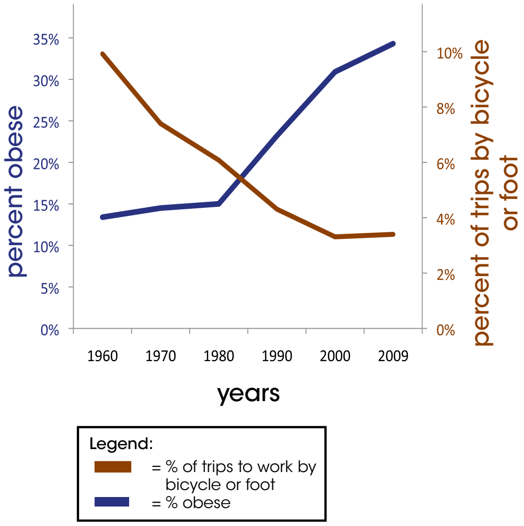
To continue to measure the potential impact of bicycling and walking levels on public health, this report analyzed data on a number of public health indicators to bike/ped mode share. Indicators include obesity and overweight levels (current and over time), physical activity levels, high blood pressure, and diabetes.

Bicycling, Walking, and Obesity

Trends over Time

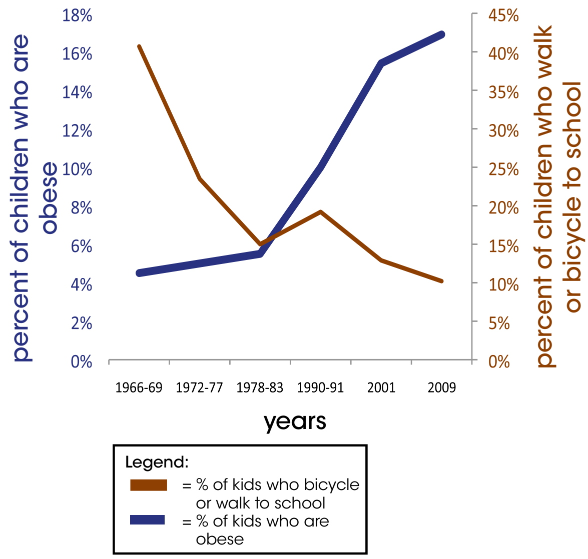
To compare rates of bicycling and walking with obesity trends, Census Journey to Work and ACS data for 1960 through 2009 were compared to overweight and obesity levels in the United States for

Change in Bicycling and Walking Rates vs. Adult Obesity Rates



Sources: Ogden and Carroll 2010, Census 1960,1970,1980,1990, 2000, ACS 2009 **Note:** bicycling was not separated from "other" modes in early Census surveys, so 1960 and 1970 levels shown are for walking only; $r = -0.93$ (bicycle + walk/overweight); $r = -0.87$ (bicycle + walk/obesity).

Trend in Obese Children vs. Rate of Bicycling and Walking to School



Sources: CDC, NHANES, McDonald 2007, Ogden and Carroll 2010, NHTS 2009 **Note:** $r = -0.70$.

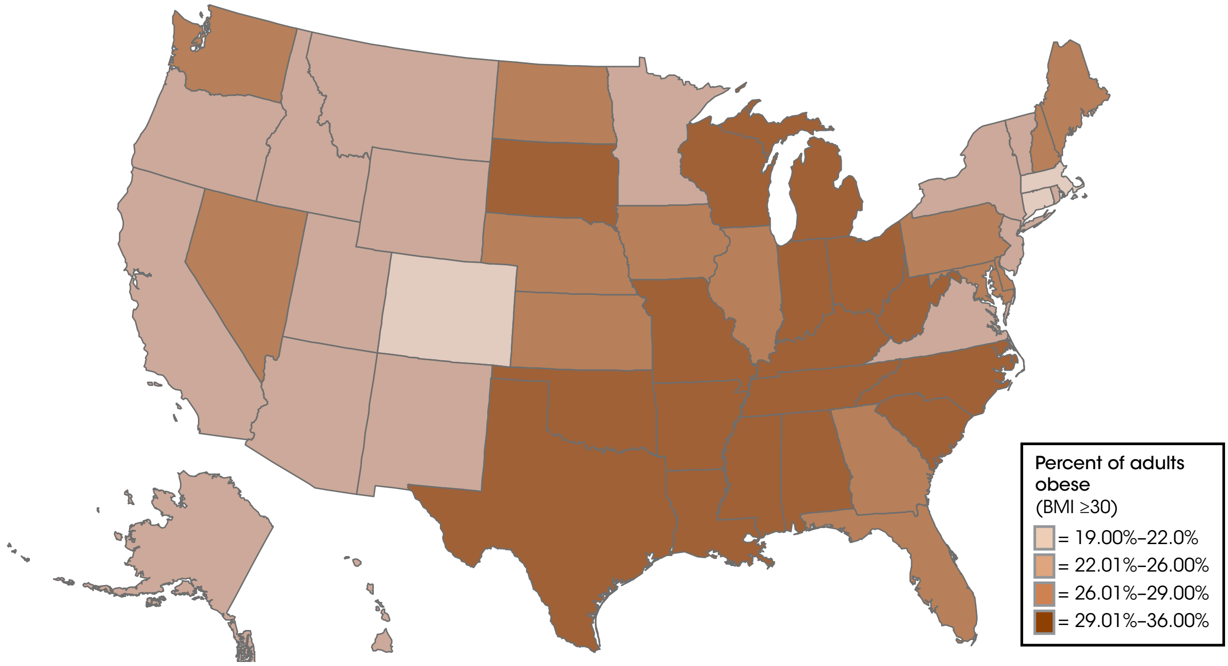
the same time period. These data show that as bicycling and walking levels have plummeted, overweight levels have steadily increased and obesity levels have soared. The decrease in bicycling and walking may be even greater since these data do not take into account any trips besides work trips (walking and bicycling to school, for example, would not be counted here). Also, bicycling was not separated from "other" modes in early Census surveys, so 1960 and 1970 levels shown are for walking only. While bicycling and walking levels fell 66% between 1960 and 2009, obesity levels increased by 156%. Although these two trends are not the only factors involved, the correlation cannot be ignored.

This report also looked at data on childhood obesity prevalence from the CDC's National Health and Nutrition Examination Survey (NHANES) and data on levels of bicycling and walking to school from the NHTS (McDonald 2007, NHTS 2009) over a similar time period. The data demonstrate a parallel trend among schoolchildren in this time period. Levels of bicycling and walking to school declined sharply while child-

(Continued page 173)

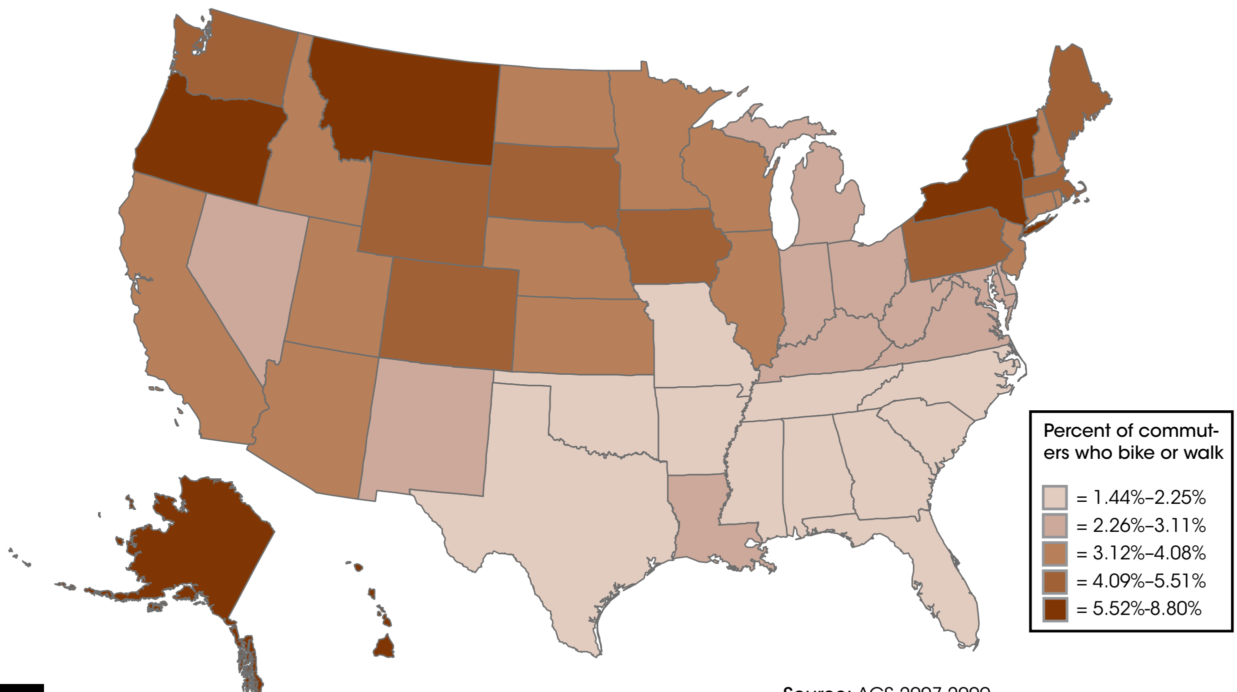
photo by Dave Delaney

Obesity Levels



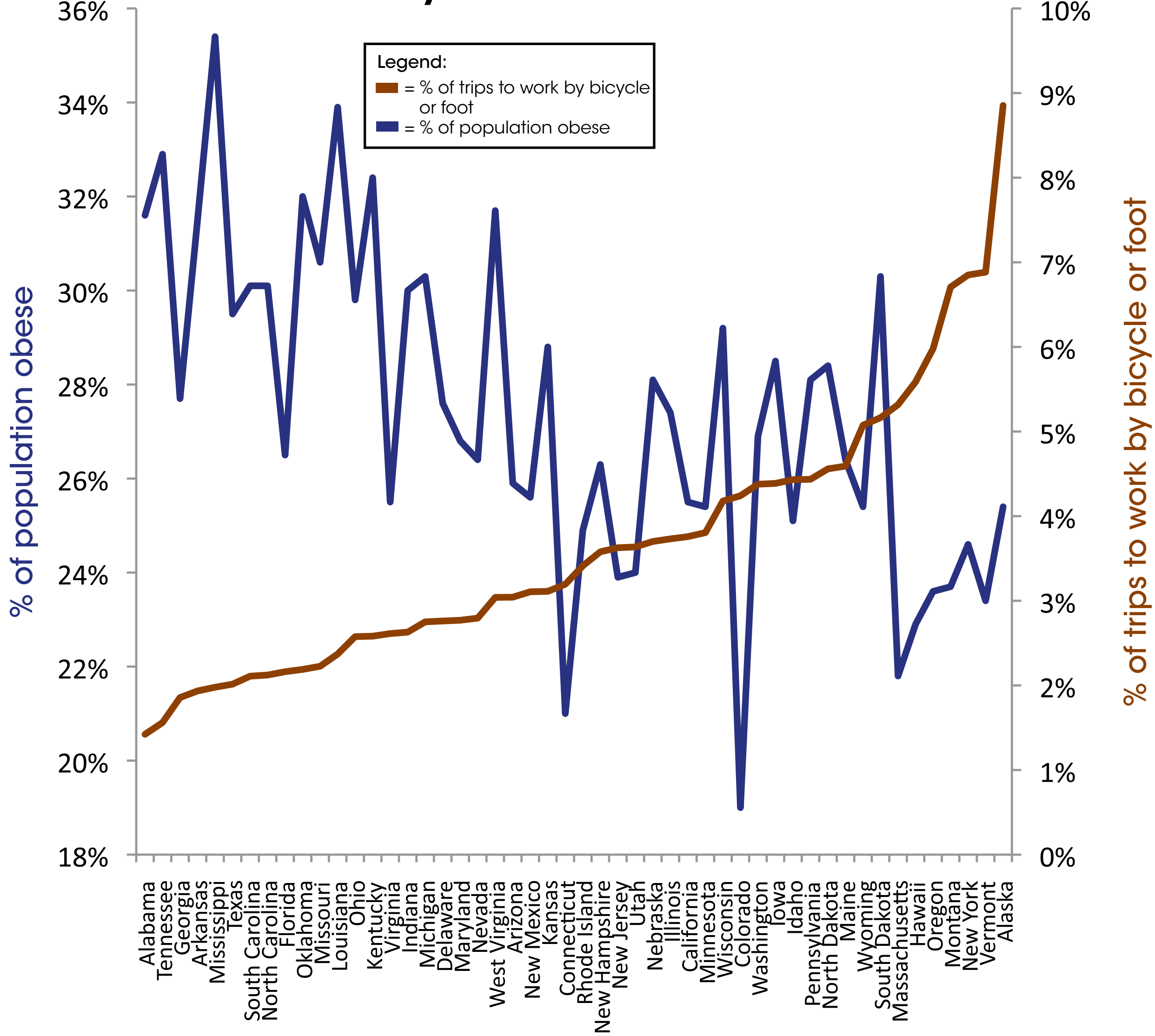
Source: BRFSS 2009

Levels of Bicycling and Walking to Work



Source: ACS 2007-2009

Comparing Bicycling and Walking to Obesity Levels in 50 States



Sources: BRFSS 2009, ACS 2009 **Note:** *r* = –0.58.

States where bicycling and walking levels are lowest have the highest levels of obesity. Data are limited to bicycling and walking trips to work, but give an idea of the comparative rates of bicycling and walking in each state.

States with higher levels of bicycling and walking average lower obesity levels.

Public Health in 50 States

State	% population overweight(1)	% population obese	% adults w/ 30+ min physical activity	% adults ever told have diabetes	% adults ever told have asthma	% adults ever told have hypertension
Alabama	68%	32%	41%	12%	8%	37%
Alaska	63%	25%	61%	6%	9%	26%
Arizona	64%	26%	51%	8%	11%	27%
Arkansas	67%	32%	47%	10%	8%	34%
California	61%	26%	51%	9%	8%	26%
Colorado	56%	19%	57%	6%	8%	22%
Connecticut	59%	21%	54%	7%	9%	27%
Delaware	64%	28%	51%	8%	9%	31%
Florida	63%	27%	46%	11%	7%	32%
Georgia	65%	28%	46%	10%	7%	31%
Hawaii	58%	23%	53%	9%	9%	30%
Idaho	61%	25%	58%	8%	8%	26%
Illinois	65%	27%	52%	8%	9%	29%
Indiana	65%	30%	48%	9%	9%	31%
Iowa	67%	29%	50%	8%	7%	28%
Kansas	65%	29%	49%	9%	9%	29%
Kentucky	67%	32%	46%	12%	10%	36%
Louisiana	68%	34%	44%	11%	6%	36%
Maine	64%	26%	56%	8%	11%	30%
Maryland	63%	27%	49%	9%	9%	29%
Massachusetts	58%	22%	53%	8%	11%	26%
Michigan	66%	30%	52%	9%	10%	30%
Minnesota	63%	25%	53%	6%	7%	22%
Mississippi	70%	35%	38%	12%	8%	37%
Missouri	66%	31%	50%	8%	10%	31%
Montana	62%	24%	59%	7%	8%	28%
Nebraska	65%	28%	51%	8%	8%	27%
Nevada	63%	26%	51%	8%	9%	28%
New Hampshire	63%	26%	53%	7%	10%	29%
New Jersey	62%	24%	48%	9%	8%	28%
New Mexico	62%	26%	53%	9%	9%	27%
New York	60%	25%	51%	9%	10%	29%
North Carolina	65%	30%	46%	10%	8%	32%
North Dakota	66%	28%	52%	8%	9%	27%
Ohio	67%	30%	49%	10%	10%	32%
Oklahoma	67%	32%	47%	11%	10%	34%
Oregon	61%	24%	57%	8%	11%	27%
Pennsylvania	64%	28%	50%	9%	9%	31%
Rhode Island	62%	25%	48%	7%	10%	30%
South Carolina	66%	30%	45%	10%	8%	33%
South Dakota	67%	30%	45%	7%	8%	30%
Tennessee	69%	33%	36%	10%	8%	33%
Texas	67%	30%	48%	9%	7%	29%
Utah	58%	24%	58%	6%	8%	23%
Vermont	58%	23%	58%	6%	10%	27%
Virginia	61%	26%	51%	8%	8%	28%
Washington	62%	27%	54%	8%	9%	28%
West Virginia	68%	32%	35%	12%	9%	38%
Wisconsin	66%	29%	53%	8%	10%	28%
Wyoming	62%	25%	57%	7%	9%	26%
Mean/Average (2)	63%	27%	51%	8%	9%	29%
Median	64%	27%	51%	8%	9%	29%
High	70%	35%	61%	12%	11%	38%
Low	56%	19%	35%	6%	6%	22%

Source: BRFSS 2009 **Notes:** (1) Percent overweight includes percent obese. (2) All averages are weighted.

Legend:
= High value
= Low value

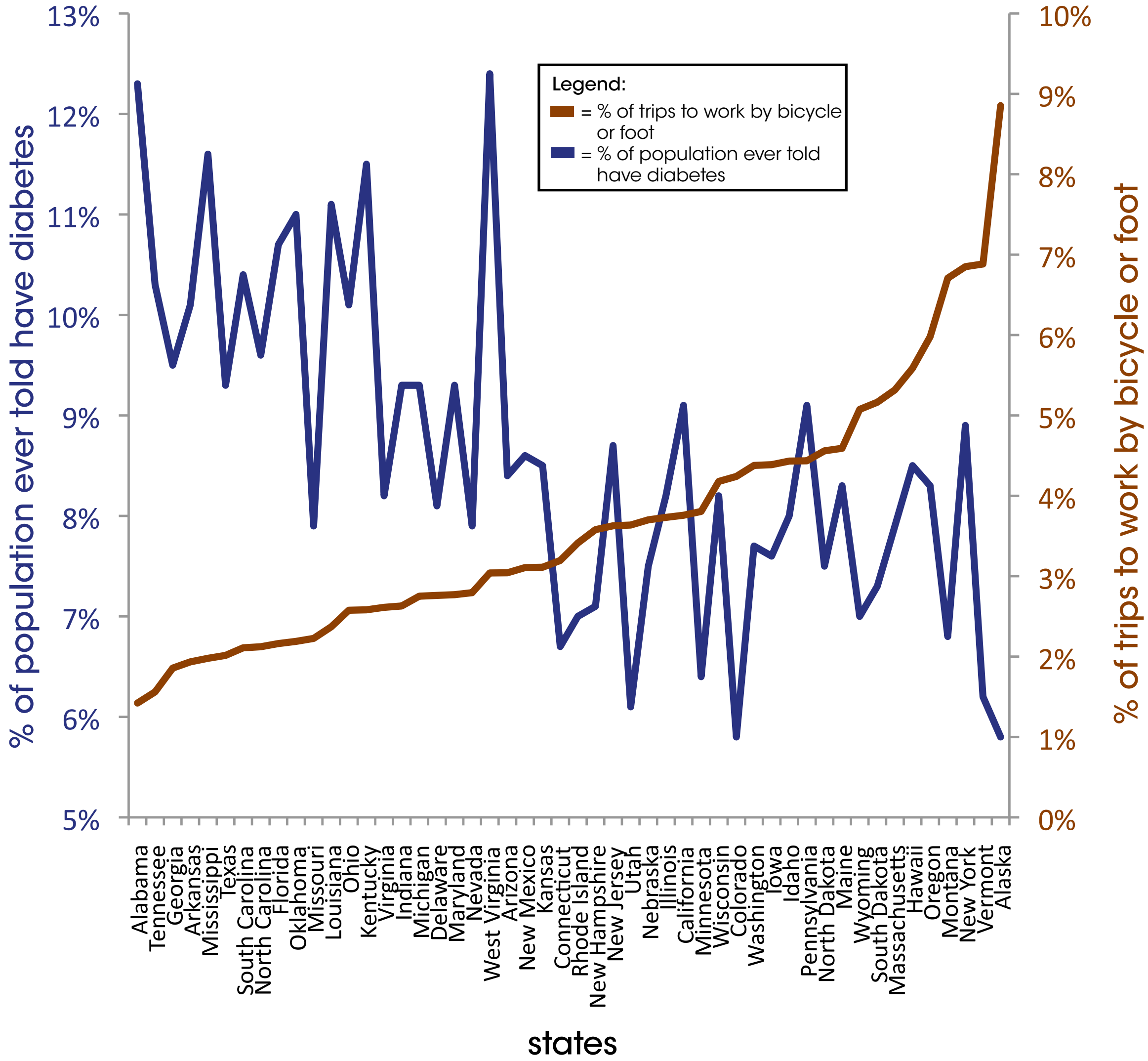
Public Health in U.S. Cities

City	% population overweight (1)	% population obese	% adults w/ 30+ min physical activity	% adults ever told have diabetes	% adults ever told have asthma	% adults ever told have hypertension
Albuquerque	60%	23%	54%	8%	9%	25%
Arlington, TX	66%	25%	54%	7%	5%	27%
Atlanta	64%	25%	46%	8%	7%	27%
Austin	61%	27%	57%	6%	7%	28%
Baltimore	63%	27%	49%	9%	11%	30%
Boston	56%	21%	52%	7%	10%	28%
Charlotte	63%	26%	48%	8%	7%	29%
Chicago	64%	28%	50%	8%	9%	28%
Cleveland	66%	29%	51%	11%	12%	35%
Colorado Springs	57%	18%	57%	5%	7%	20%
Columbus	63%	29%	48%	11%	7%	28%
Dallas	63%	26%	47%	8%	7%	25%
Denver	56%	19%	56%	5%	8%	24%
Detroit	68%	34%	48%	11%	10%	32%
El Paso	64%	28%	50%	12%	7%	29%
Fort Worth	66%	25%	54%	7%	5%	27%
Honolulu	57%	22%	52%	9%	9%	31%
Houston	64%	27%	46%	8%	6%	27%
Indianapolis	66%	28%	48%	8%	10%	29%
Jacksonville	61%	25%	49%	10%	9%	27%
Kansas City, MO	62%	27%	49%	8%	9%	27%
Las Vegas	63%	28%	49%	8%	9%	27%
Long Beach	61%	26%	46%	10%	7%	26%
Los Angeles	61%	26%	46%	10%	7%	26%
Louisville	65%	34%	46%	12%	10%	36%
Memphis	72%	34%	38%	11%	7%	38%
Mesa	65%	25%	50%	8%	12%	25%
Miami	62%	24%	43%	10%	5%	32%
Milwaukee	61%	24%	53%	8%	8%	28%
Minneapolis	62%	24%	55%	6%	8%	20%
Nashville	66%	27%	40%	7%	7%	26%
New Orleans	64%	29%	44%	10%	5%	36%
New York	56%	22%	47%	9%	9%	26%
Oakland	57%	22%	55%	8%	10%	25%
Oklahoma City	66%	29%	46%	10%	10%	31%
Omaha	62%	27%	52%	7%	6%	26%
Philadelphia	61%	26%	47%	9%	10%	29%
Phoenix	65%	25%	50%	8%	12%	25%
Portland, OR	60%	24%	55%	7%	10%	25%
Raleigh	64%	26%	45%	6%	7%	28%
Sacramento	62%	26%	53%	8%	10%	25%
San Antonio	64%	25%	50%	8%	8%	28%
San Diego	59%	22%	58%	8%	7%	24%
San Francisco	50%	17%	54%	7%	8%	23%
San Jose	53%	21%	47%	9%	5%	22%
Seattle	59%	24%	52%	7%	8%	26%
Tucson	60%	28%	52%	9%	11%	27%
Tulsa	66%	31%	46%	10%	10%	35%
Virginia Beach	65%	24%	53%	8%	9%	25%
Washington, DC	59%	26%	50%	8%	8%	26%
Mean/Average (2)	61%	25%	49%	9%	8%	27%
Median	62%	26%	50%	8%	8%	27%
High	72%	34%	58%	12%	12%	38%
Low	50%	17%	38%	5%	5%	20%

Source: BRFSS 2009 **Notes:** Data unavailable for Fresno. (1) Percent overweight includes percent obese. (2) All averages are weighted.

Legend:
= High value
= Low value

Comparing Bicycling and Walking to Diabetes Rates in 50 States

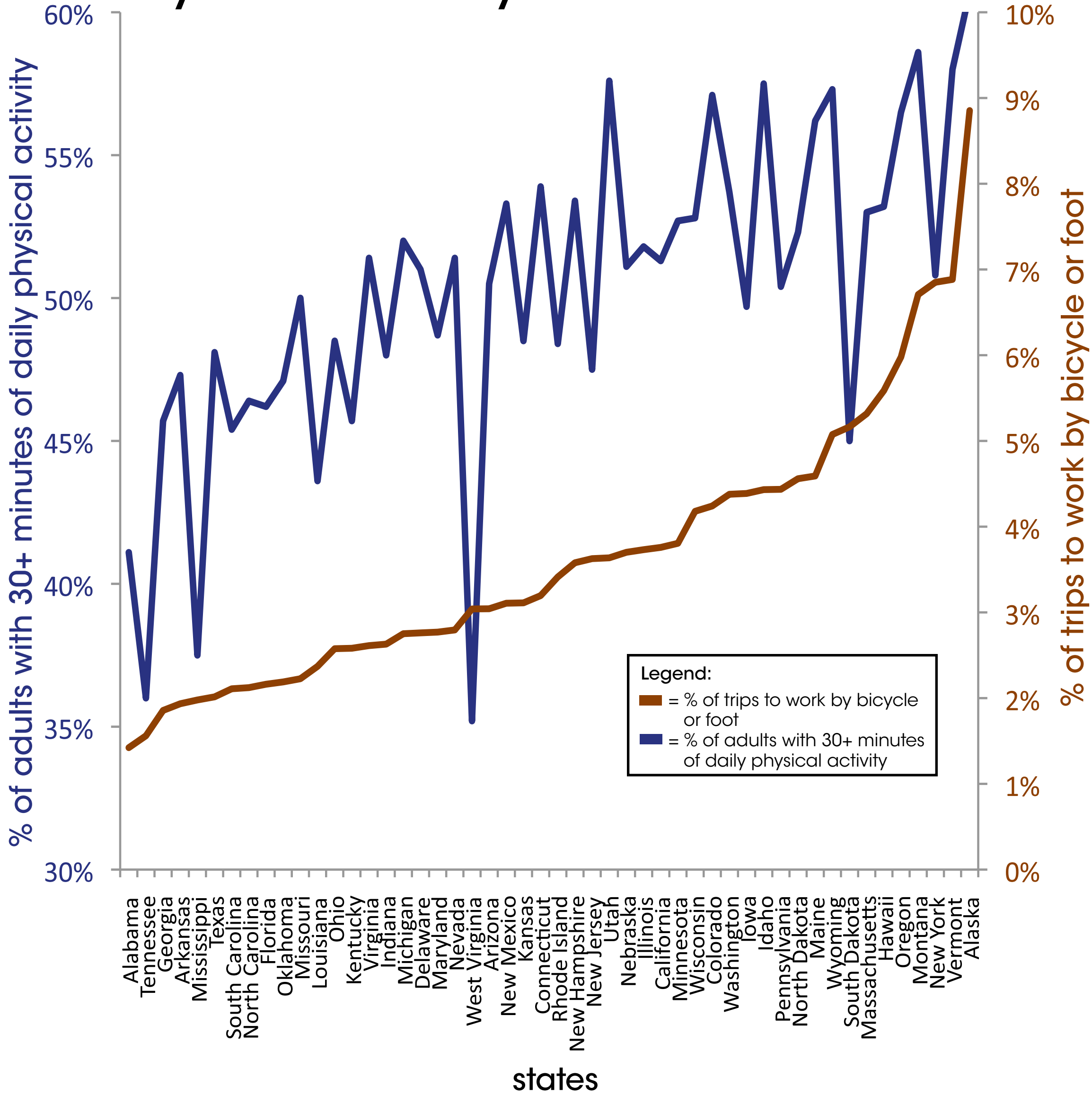


Sources: BRFSS 2009, ACS 2009 **Note:** *r* = –0.63.

Diabetes rates are lowest among states with high levels of bicycling and walking.

Data suggest a negative correlation exists between rates of diabetes and levels of bicycling and walking ($r = -0.63$). According to data from BRFSS and ACS 2009, diabetes rates are highest among states with low levels of bicycling and walking.

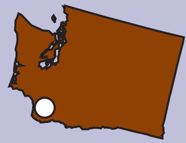
Comparing Bicycling and Walking to Physical Activity Rates in 50 States



Sources: BRFSS 2009, ACS 2009 **Note:** *r* = 0.68.

A strong positive correlation exists between levels of adults with 30-plus minutes of daily physical activity and levels of bicycling and walking to work. Data indicate a positive relationship ($r = 0.68$) between the two, suggesting that bicycling and walking to work help populations meet recommended levels of physical activity.

States with higher levels of bicycling and walking have higher levels of physical activity.



CLOSER LOOK

Clark County, WA's Health Impact Assessment

by Brendon Haggerty, Clark County Public Health, Clark County, WA

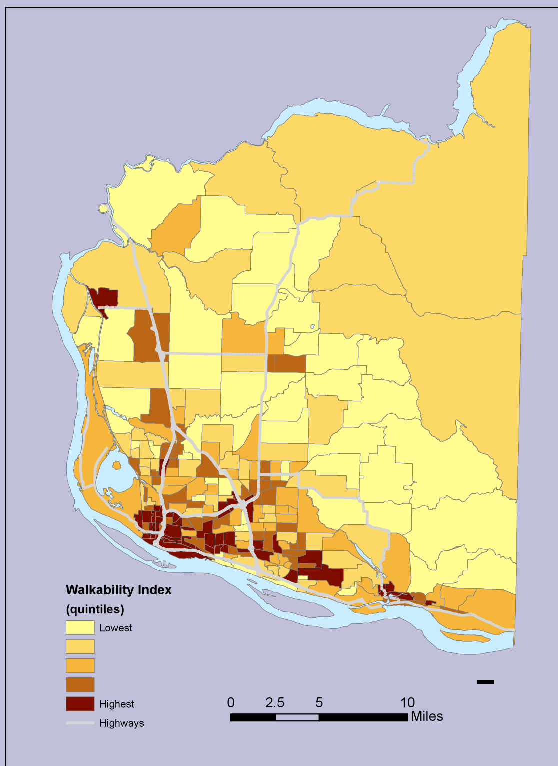
Obesity, an underlying factor for several chronic diseases, has become epidemic in the United States, and, as such, is a central challenge for public health. Reversing this epidemic will require many actions, and among them is addressing the role of the built environment in contributing to obesity. To do this, local agencies are increasingly using health impact assessment (HIA) to evaluate the potential health effects of a project or policy. Clark County Public Health (CCPH) is one example of a local agency that has used HIA to integrate health concerns, such as obesity, into public decisions related to the built environment.

In December 2010, Clark County adopted its first Bicycle and Pedestrian Master Plan. With the support of the Robert Wood Johnson Foundation, CCPH partnered with the County's Community Planning Department to conduct an HIA on the plan. The HIA focused on prioritizing projects, programs, and policies to maximize the health benefits of physical activity. The health benefits of walking and cycling are obvious, but CCPH wanted to prioritize projects in a way that benefited populations more vulnerable to poor health outcomes, including low-income households, racial and ethnic minorities, youth, and adults over age 65.

After performing an extensive geographic information system (GIS) analysis and identifying existing research findings on physical activity and obesity, CCPH recommended programs, policies, and implementation strategies designed to increase opportunities for physical activity and reduce obesity. CCPH calculated a walkability index (see map) and overlaid it with demographic data to determine where projects could best serve populations in need of increased opportunities for physical activity. Among the key recommendations were:

- Include low-speed roadway designs as bicycle and pedestrian projects
- Implement a variety of innovative bikeway facility types
- Integrate policies to improve street connectivity, urban design, land use mix, and residential density

Clark County HIA Walkability Index



- Create policies to increase bicycle and pedestrian access to nutritious food
- Include health and equity in project evaluation criteria
- Prioritize improvements in low socioeconomic status neighborhoods

As a result of the HIA, the plan includes policies to address health and equity by integrating health concerns in the selection of priority improvements. Of a possible score of 100, the project prioritization scheme awarded 20 points based on the potential to increase physical activity and health equity. Accordingly, planned improvements are heavily weighted toward low-income areas, which will see about half of all proposed improvements (see table below).

Block Group Median Income	% of Proposed Sidewalk Miles	% of Proposed Bikeway Miles
Lowest 1/3	51%	45%
Middle 1/3	17%	25%
Highest 1/3	32%	30%

In follow-up interviews, participants in the planning process said that the HIA helped to frame their decisions and contributed to their understanding of the importance of active transportation in promoting physical activity. Elected officials and planners stated that information from the HIA helped them communicate with the public about the need for bicycle and pedestrian planning. Partially as a result of a positive experience with this HIA, other HIAs have followed, and the county is adding a chapter on health to its comprehensive plan. Conducting an HIA helped the county begin to consider health in all policies, and the HIA was ultimately awarded a Model Practice Award from the National Association of County and City Health Officials.

The HIA and supporting documents are available at:
www.co.clark.wa.us/public-health/reports/facts.html.

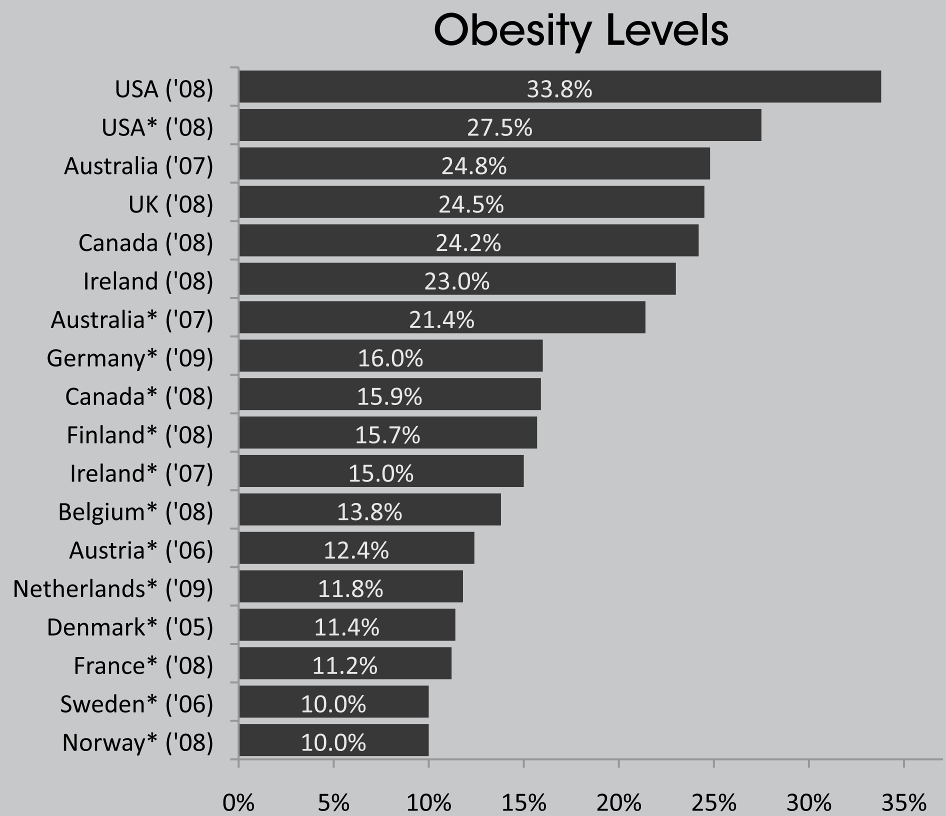
Mt. Hood is the backdrop for this bicycle trip in Clark County, WA.
Photo courtesy of Laurie Lebowsky, Clark County Community Planning

Looking Outside the Borders

Obesity Levels in Developed Nations

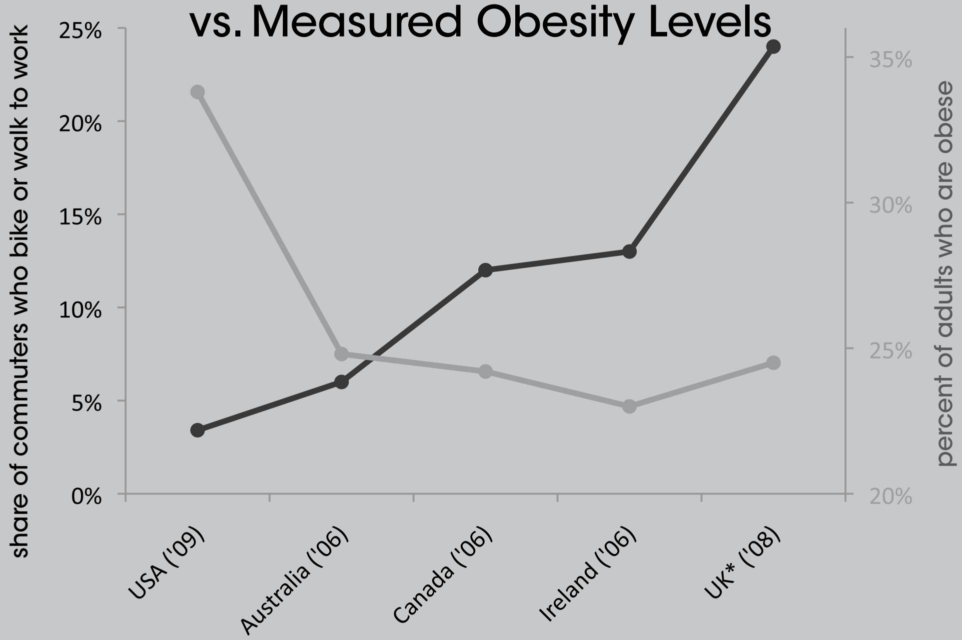
According to data from Organisation for Economic Co-operation and Development, the United States has the highest obesity levels among developed countries. A 2010 study by Pucher and Buehler show the United States also ranks at the bottom for levels of bicycling and walking when compared to international peers. These international data further suggest a correlation between lower levels of bicycling and walking and higher levels of obesity.

> ...international data further suggest a correlation between lower levels of bicycling and walking and higher levels of obesity.



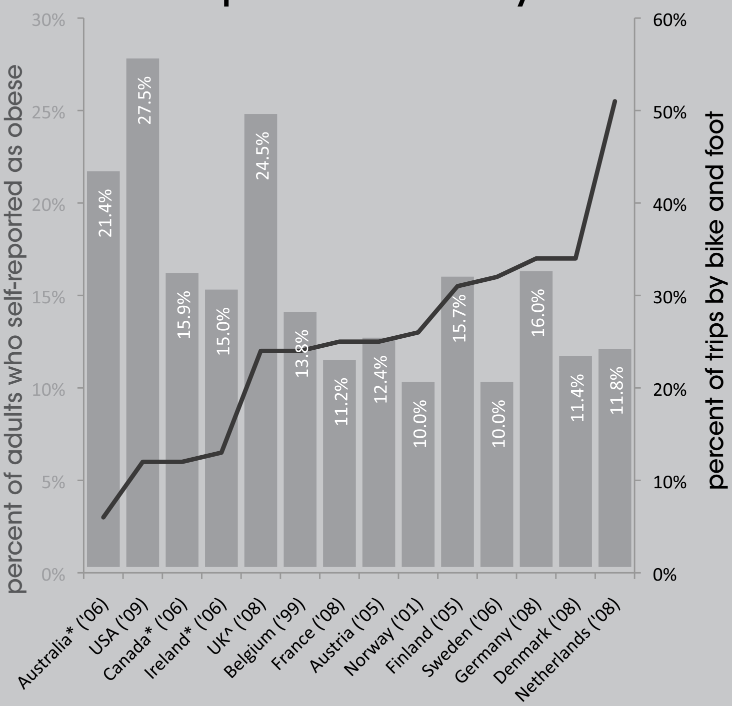
Source: Organisation for Economic Co-operation and Development (OECD) 2010 (www.oecd.org/health/healthdata) **Note:** * = obesity levels are self-reported.

Levels of Bicycling and Walking to Work vs. Measured Obesity Levels



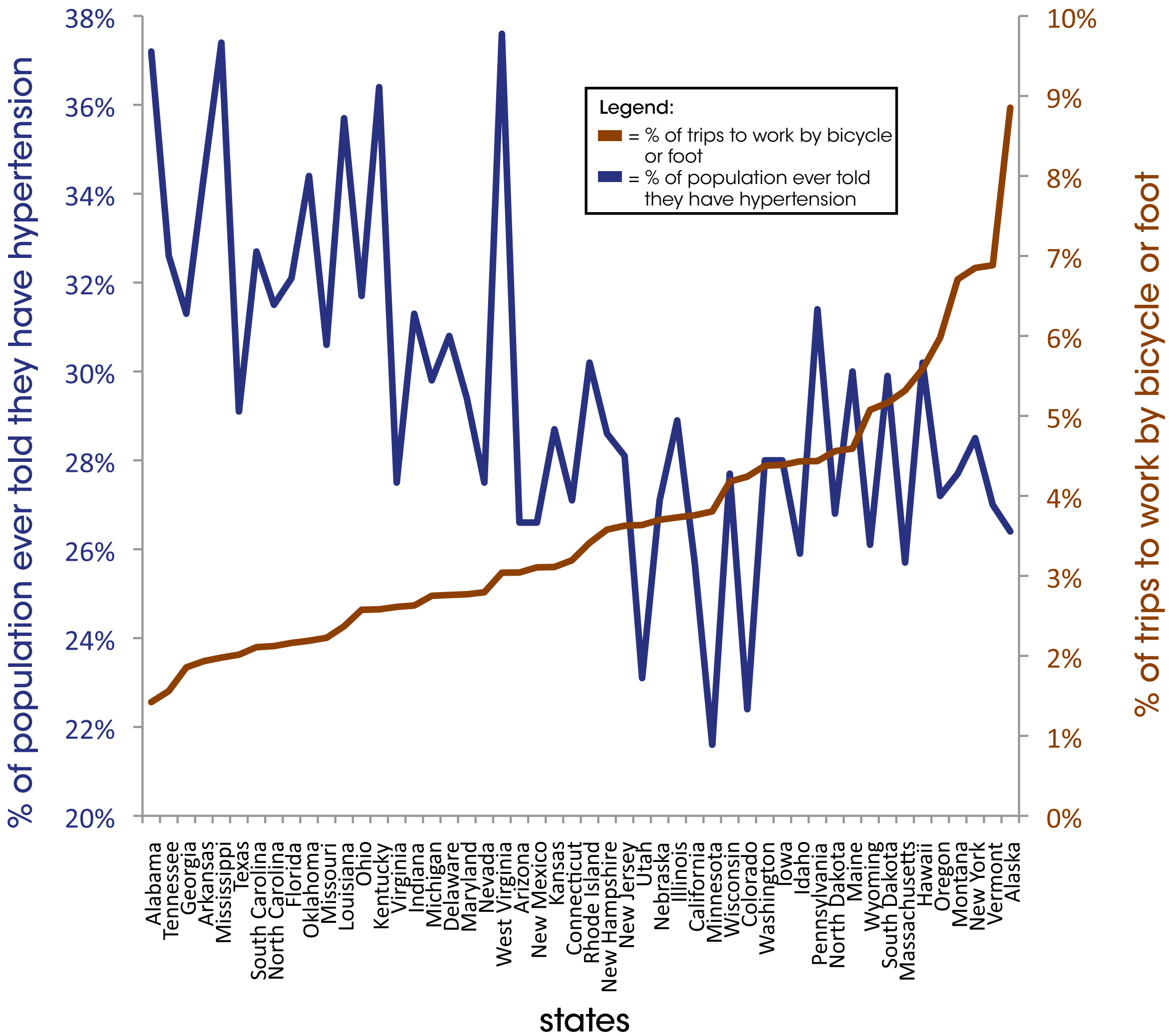
Source: J. Pucher and R. Buehler, 2010. "Walking and Cycling for Healthy Cities," *Built Environment 36*(4), pp. 391-414, and Organisation for Economic Co-operation and Development (OECD) 2010 (www.oecd.org/health/healthdata), * = UK bicycling and walking data is for all trips, not just work trips. **Note:** *r* = -0.54.

Levels of Bicycling and Walking vs. Self-Reported Obesity Levels



Source: J. Pucher and R. Buehler, 2010. "Walking and Cycling for Healthy Cities," *Built Environment 36*(4), pp. 391-414, and Organisation for Economic Co-operation and Development (OECD) 2010 (www.oecd.org/health/healthdata). **Notes:** *r* = -0.58 * = bicycling and walking data represents share of commuters who bike and walk to work and does not represent all trips ^ = UK obesity data is measured, not self-reported. Years in parentheses are for percent of trips by bike and foot data. For years of obesity data, reference chart on previous page.

Comparing Bicycling and Walking to High Blood Pressure Rates in 50 States



Sources: BRFSS 2009, ACS 2009 **Note:** *r* = −0.52.

States with higher levels of bicycling and walking average lower prevalence of high blood pressure.

Data from BRFSS and ACS suggest a negative correlation between levels of high blood pressure and bicycling and walking ($r = -0.52$). This relationship is in line with other results indicating a similar negative correlation between bicycling and walking levels and prevalence of obesity and diabetes.

hood obesity levels sharply increased. During the period between 1966 and 2009, the number of children who biked or walked to school fell 75%, while the percentage of obese children rose 276% (McDonald 2007, NHTS 2009).

Comparing Obesity Levels to Bicycling and Walking

The Alliance used ACS data on bicycling and walking to work, and Behavioral Risk Factor Surveillance System (BRFSS) data on obesity levels across states to compare current levels of obesity with bicycling and walking. The data indicate that states with the highest levels of bicycling and walking to work have lower levels of obesity on average.

Other Health Indicators

This report also compared rates of bicycling and walking to work to other health indicators including levels for physical activity, rates of high blood pressure, and diabetes. Data suggest a strong positive correlation between rates of bicycling and walking and levels of physical activity. States with the highest levels of bicycling and walking have a greater percentage of the population meeting the recommended 30-plus minutes a day of physical activity. A negative correlation exists between rates of bicycling and walking and high blood pressure and diabetes. States with higher levels of bicycling and walking have lower levels of both diabetes and high blood pressure on average.

Photo courtesy of Transportation for America

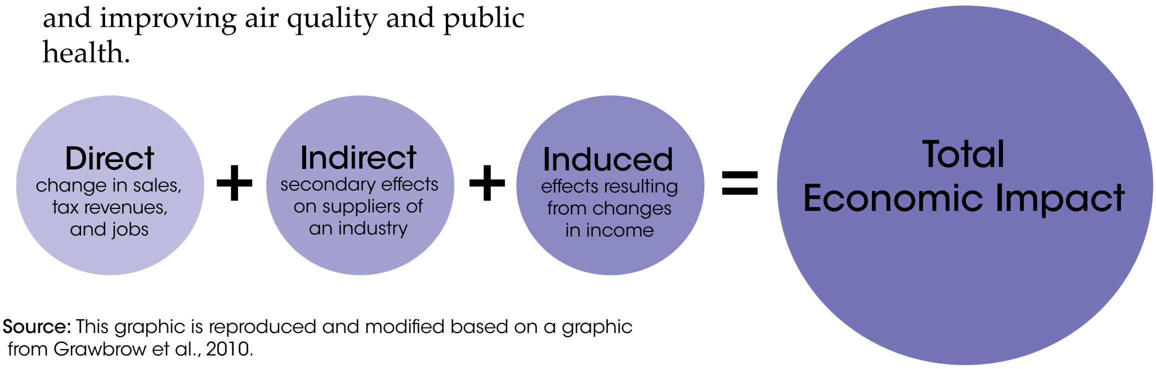
Photo by Stephen Lee Davis

9: ECONOMIC BENEFITS

As economic recession has hit almost every level of our society over the last few years, active transportation has emerged as a promising sector for growth and revitalization. Although research on the economic impact of bicycling and walking is limited, recent studies have shown that communities that invest in these modes have higher property values, create new jobs, and attract tourists. In addition, these communities save money by decreasing traffic congestion and commute times and improving air quality and public health.

Estimating Economic Impact

There are many ways the economic impact of bicycling and walking has been and could be measured. Some simple methods include surveys to trail users or event participants that ask them about their spending relating to bicycling or walking. Others involve more complex modeling. The input-output model estimates the complete impact



Source: This graphic is reproduced and modified based on a graphic from Grawbrow et al., 2010.

of bicycling or walking on the economy by including the direct, indirect, and induced effects of the industry or infrastructure.

For example, if you wanted to measure the economic impact of a specific trail, you would first quantify the direct impact. This includes changes in sales, tax revenues, and jobs directly attributed to the trail. Examples might include sales at a bike shop located on the trail by trail users, food purchases by trail users, and hotel accommodations by tourists whose primary reason for coming to the area was to use the trail.

Next, quantifying indirect impact includes the secondary effects on suppliers to the industries directly affected. For example, this would include businesses such as the dairies and creameries who supply ice cream to the snack stands that trail users patronize.

Lastly, induced impact accounts for the spending of income by people whose employment is dependent on the trail. This model gives a comprehensive look at how money flows through the economy because of the trail.

A summary of studies estimating the economic impact of bicycling and walking can be found on page 177. Studies vary widely in their scope, methodology, and estimates.

An Economic Boost

Years of planning and building streets for cars has left many communities severely lacking for bicycle and pedestrian infrastructure. Building new facilities for bicycling and walking can be a boost for the economy. In 2010 Bikes Belong developed a series of 10 case studies on U.S. bicycle and pedestrian facilities built, at least in part, with federal transportation dollars. Each of the projects profiled created between 218 and 1,050 new construction jobs. Other impacts on local economies included rising property values, increased business at local establishments, and savings from reduced traffic congestion.

(Continued page 178)

Job Creation from Bike/Ped Projects

Location	Project	Jobs Created in Construction	Other Economic Impacts
Baltimore, MD	Average project	11-14 jobs/$1 million spent	Not quantified.
Charleston, SC	Wonders Way Path/Ravenel Bridge	525+	Three local bike shops reported significantly more customers after path was built
Chicago, IL	McDonald's Cycle Center	unknown	Center employs 30+ people
Jackson Hole, WY	Grand Teton National Park Pathways	525	500,000 visitor trips/year
Minneapolis, MN	Midtown Greenway	700	Home values increase $510 for every 400 meters closer they are to off-street facilities like the Greenway
New York, NY	Williamsburg Bridge	595	5,200 bike users/day. Average car driver in Manhattan causes 3.26 hours of delays to other drivers, equivalent to $160/day
Portland, OR	Vera Katz Eastbank Esplanade	1,050	Adjacent building was renovated to provide space for 185 jobs, 100 of them new to the city
San Francisco, CA	Valencia Street Redesign	218	2/3 merchants said the redesign improved business

Sources: Bikes Belong 2011, Garrett-Peltier 2010

Photo courtesy of Frank Chan, San Francisco Bicycle Coalition

Overview of Bike/Ped Economic Impact Studies

State	Location/scope	Economic Impact	Study Link
California	San Francisco: Valencia Street bike lane	Sixty-six percent of merchants believe that the bike lanes have had a generally positive impact on their business and/or sales, and the same percentage would support more traffic calming on Valencia Street (2003).	http://www.emilydrennen.org/TrafficCalming_full.pdf
Colorado	Statewide	Total economic benefit from bicycling in Colorado is over $1 billion annually; 1,316 full time and 7,500 seasonal bicycle manufacturing, retail, and tourism jobs in the state (1999).	http://atfiles.org/files/pdf/CObikeEcon.pdf
Florida, California, and Iowa	Three rail trails	$1.2 to $1.9 million annually in economic activity and pumps $294,000-$630,000 annually into local trail communities. (1992).	http://www.brucefreemanrailtrail.org/pdf/1_Exec_summ__contents.pdf
Maine	Statewide bicycle tourism	Bicycle tourists spend $36.3 million annually with a total economic impact of $66.8 million each year (2001).	http://www.maine.gov/mdot/opt/pdf/biketourismexecsumm.pdf
Maryland/ Pennsylvania	Great Allegheny Passage	Trail attributed revenue in 2007 was $32,614,703 and it was projected that businesses distributed $6,273,927 in wages. Despite the tough economic times, in 2008 these figures increased to projected receipts and wages of $40,677,299 and $7,500,798, respectively (2009).	http://www.adventurecycling.org/routes/nbrn/resourcespage/GAPeconomicImpactStudy200809.pdf
Maryland	Baltimore	Pedestrian and bicycle infrastructure projects create 11-14 jobs per $1 million of spending (2010).	http://www.americabikes.org/Documents/PERI_Case_Study-Baltimore.pdf
Minnesota	Trails throughout the state	Statewide trail spending of $2,422 million was estimated to produce $2,953 million in gross output (total sales of local businesses including indirect and induced effects but subtracting imports). This contributed $1,542 million to gross state product (GSP). Some 30,900 full-time and part-time jobs were supported by trail spending in various regions. Employee compensation from these jobs reached some $864 million. State and local revenues from all taxes, fees, and other sources amounted to $206 million (2009).	http://www.tourism.umn.edu/prod/groups/cfans/@pub/@cfans/@tourism/documents/asset/cfans_asset_167538.pdf
New York	Eight trails throughout New York	Trail users spent average of $28.90/visit, $200 on equipment related to use of the trail, Total economic impact of just one of the eight trails studied was estimated to be $2 million annually, supporting roughly 40 FTE (Full Time Equivalent) jobs within the community. This estimate does not include increased tax revenues to the local community (2010).	http://nysparks.com/recreation/trails/statewide-plans.aspx.; Statewide Trails Plan. Appendix C – Every Mile Counts – An Analysis of the 2008 Trail User Surveys.
North Carolina	Outer Banks bicycle tourism	A conservative estimate of the annual economic impact of bicycle tourists is $60 million, almost nine times as much as the one-time expenditure of public funds used to construct special bicycle facilities in the region. 1,400 jobs are created or supported annually with the expenditures made by bicycle tourists (2004).	http://www.ncdot.gov/bikeped/download/bikeped_research_EIAfulltechreport.pdf
Oregon	Portland	Health cost savings attributable to new bicycle infrastructure and programs will be between $388 and $594 million by 2040. The savings in value of statistical lives is between $7 and $12 billion (2011).	http://www.portland-mercury.com/images/

Overview of Bike/Ped Economic Impact Studies (Continued)

Pennsylvania	Pine Creek Rail Trail	82% of survey respondents on trail had purchased "hard goods" (bikes, bike accessories, clothing, etc.) at an average of $354/user. 86% had purchased "soft goods" (water, soda, candy, ice cream, etc.) at an average of $30/user. 57% of users were tourists who reported staying in hotels that averaged $69/night (2006).	http://www.railstotrails.org/resources/documents/resource_docs/RTC_PineCreekGuide_web.pdf
Pennsylvania	Heritage Rail Trail	85% of trail users surveyed purchased some form of "hard goods" (defined as bikes; bike accessories; auto accessories; running, walking, hiking shoes or clothing) in conjunction with their use of the trail. The average spending of those who provided spending data was $367. Nearly 72% had purchased "soft goods" (water, soda, candy, ice cream, lunches, etc.) averaging $12.66. 12% of users said their use of trail involved an overnight hotel stay with an average cost of $51 (2007).	http://www.yorkcountyparks.org/PDF/2007%20Rail%20Trail%20User%20Survey%20Report%20VERSION%204.1.pdf
Virginia	Virginia Creeper Trail	Economic impact of nonlocals was $2.2 million. Total economic impact was $2.5 million/year (2004).	http://atfiles.org/files/pdf/VAC-study04.pdf
Wisconsin	Statewide comprehensive	This study estimates the economic impact of bicycle recreation and tourism in Wisconsin to be $924,211,000, and the total potential value of health benefits from reducing short car trips and increasing bicycle trips to total $409,944,167 (2010).	http://www.bfw.org/uploads/media/Valuing_Bicycling_in_Wisconsin_Final_Report_January_2010[1].pdf

A December 2010 study used the input-output model to estimate the number of jobs created by pedestrian and bicycle infrastructure projects and other road projects. According to the author, "pedestrian and bicycle infrastructure projects create 11-14 jobs per $1 million of spending while road infrastructure projects create approximately 7 jobs per $1 million of expenditures" (Garrett-Peltier 2010).

Lasting Impact

After the initial economic boost from construction, pedestrian and bicycle infrastructure has lasting effects on local economies.

Events and Tourism

It has long been recognized that facilities like rail trails and safe places to bike and walk attract tourists. Local communities now vie for "Bicycle Friendly Community" and "Walking Friendly Community" designation and communities with this designation report it is good for business (Maus 2006).

Numerous studies and papers have looked at the impact of bicycling on tourism. A 2004 study (detailed on page 181) found the annual economic impact of bicycle tourists to North Carolina's Outer Banks was $60 million. In addition, 1,400 jobs were created or sustained annually because of these tourists.

Bicycling and walking events can also stimulate local economies. Iowa's RAGBRAI, a weeklong bicycle ride across the state, contributed $16.5 million in direct spending and supported 362 jobs in the state (Lankford 2008). The Tour of Missouri, a professional cycling race, estimated their direct


Tour de Brooklyn 2009 Photo by Lois Bielefeld


Photo by Maguis and David

Chilly Hilly Participants Arrival by Ferry, Photo by Kimball Andrew Schmidt

economic impact over 3 years at over $80 million with tax revenues at $38 million. Charity Walk in Hawaii drew more than 9,100 walkers who raised over $964,500 for local charities in 2010. In Wisconsin, the estimated impact of two annual bicycling events was between $3.7 and $6.2 million in 2004 dollars. Other estimates of the impact of bicycling and walking events can be found in the table on page 180.

Good for Business

A 2009 study found that bicyclists on Minnesota trails spend $2.4 billion annually producing $2.9 billion in gross output (sales by local businesses including indirect and induced effects and subtracting imports). This added $1.5 billion to the gross state product (GSP). An estimated 30,900 full-time and part-time jobs, with roughly $864 million in compensation, were supported by trail spending. The state and local governments brought in $206 million in revenue from taxes , fees, and other sources attributable to the trails (Venegas 2009).

A 1992 study of three rail trails found that trail use generated roughly $1.5 million in economic activity annually and upward of $300,000 in local communities (Moore 1992). A 2004 study of the Virginia Creeper Trail found that every trail visitor generated between $24 and $38 per visit. The study estimated that trail visitors contributed $1.2 million annually to the local economy (Bowker 2004).

One recent study of the Great Allegheny Passage, a 132-mile system of hiking and biking trails connecting McKeesport, PA (near Pittsburgh, PA) to Cumberland, MD, suggests that bicycling and walking remain thriving economic sectors even during recession.

Economic Impact of Bike/Ped Events

Location	Event	Event Description	Estimated Economic Impacts
Florida (2)	Bike Florida	Week-long bicycle camping tour	75% of participants (1,000) were from out of state. According to a survey, 36% stayed in hotels, 95% dined at restaurants, 66% shopped for groceries, and 55% purchased clothing.
Hawaii (5)	Charity Walk	A statewide annual charity walk that occurs simultaneously on Oahu, Maui, the Big Island, and Kauai.	More than 9,100 walkers raised over $964,500 for over 220 local charities.
Iowa (7)	RAGBRAI	Annual weeklong bike ride across state of Iowa attracting roughly 9,500 bicyclists.	RAGBRAI participants' expenditures had a direct economic impact of $16.5 million in direct sales, $10.4 million in value added/ income, and supported 362 jobs in the state.
Michigan (4)	Midwest Tandem Bike Rally	An annual recreational bicycling event staged in a different location each year over a weekend. In 1999, Midland was selected as the site and attracted 550 tandem bicycle teams to the event held on the Pere Marquette Rail-Trail.	$260,000 of direct spending by travel parties in Michigan. 25% of the events' participants said the Pere Marquette Rail-Trail was the primary draw to ride participation.
Michigan (4)	Michigander	Recreational bicycle special events held on the Pere Marquette Rail-Trail in Midland County Michigan during the summer of 1999.	$207,000 of direct spending by travel parties in Michigan. 25% of the events' participants said the Pere Marquette Rail-Trail was the primary draw to ride participation.
Missouri (1)	Tour of Missouri	The United State's second biggest professional cycling race event.	>$80 million in direct economic impact over three years; tax revenues estimated at $38 million.
Ohio (6)	Harbin Park Cyclocross Race	Cyclocross race.	Estimated impact of $200,000 to local community.
Wisconsin (3)	Aids Network ACT II	Bicycle fundraising ride with 110 riders in 2004.	Total estimated impact = $342,400 Amount fundraised = $274,000 88% of funds raised served Aids Network clients throughout Wisconsin.
Wisconsin (3)	GRABAAWR and SAGBRAW	The Great Annual Bicycle Adventure Along the Wisconsin River (GRABAAWR) had 900 riders in 2004, and Sprocket's Annual Great Bicycle Ride Across Wisconsin (SAGBRAW) had 1,140 riders in 2004.	The economic impact of the combined events was between $3.7 million and $6.2 million in 2004. These figures were based on the event budgets, which total about $350,000 a year, plus actual spending by the participants in the two cross-state bicycling events, and appropriate multipliers to estimate indirect and induced economic effects. GRABAAWR participants reported spending an average of $60/ day, while SAGBRAW participants reported spending an average of $57/day.

Sources: (1) Tour of Missouri 2010 (2) Bike Florida 2010 (3) Grabrow et al. (4) Nelson et al., 2001(5) Charity Walk 2011 (6) Liberles 2010 (7) Lankford et al., 2008

Trail-attributed revenue in 2007 was $32,614,703 and it was projected that businesses distributed $6,273,927 in wages. Despite the tough economic times, in 2008 these figures actually increased to projected receipts and wages of $40,677,299 and $7,500,798, respectively (Campos 2009).

Business owners are aware of the impact new bicycle facilities have on business. A survey of San Francisco Valencia Street businesses, 4 years after bike lanes were installed on the street found that nearly two-thirds of merchants (65%) thought the bike lanes had an overall positive impact on their business and/or sales. Thirty percent believed the bike lanes had no impact. Just one merchant said the lanes had a "slightly negative" impact on business.

Similarly, a 2006 survey of over 100 Portland, OR businesses asked if Portland's reputation for bike-friendliness is good for business. Eighty-two percent

CLOSER LOOK

North Carolina Attracts Bicycle Tourists

A 2004 study by the Institute for Transportation Research and Education (ITRE) at North Carolina State University surveyed bicyclists riding on the bicycle facilities in the northern Outer Banks region. The study made a "conservative" estimate of the annual economic impact of bicycle tourists as $60 million, with 1,400 jobs created/supported per year. According to the study, "This compares favorably to the estimated $6.7 million of federal, state and local funds used to construct the special bicycle facilities in the area." Additional findings included:

- "The quality of bicycling in the region had a positive impact on respondents' vacation planning with 43% reporting that bicycling was an important factor in their decision to come to the area, 53% reported bicycling as a strong influence in their decision to return in the future, and 12% reported staying 3 to 4 days longer to bicycle in the area.
- Nearly two-thirds of respondents indicated that riding on bicycle facilities made them feel safer.
- Over three-fourths of all survey respondents indicated that additional bicycle paths, paved shoulders, and bike lanes should be built.
- Nine out of ten survey respondents strongly agreed that state and/or federal tax dollars should be used to build more bicycle facilities."

Sources: Lawrie et al, 2004

"Nine out of ten survey respondents strongly agreed that state and/or federal tax dollars should be used to build more bicycle facilities."

Tourists exiting ferry while bicycling North Carolina's Outer Banks. Photo courtesy of North Carolina DOT.

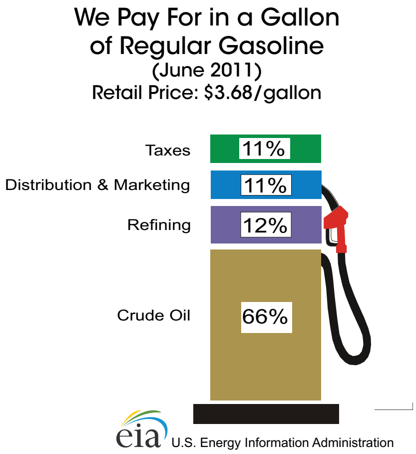
responded "Absolutely/yes" and only 3% responded "no."

Property Values

Numerous studies have examined the effect of proximity to trails and other bicycling and walking facilities on property values. A study of recreational trails in Omaha, NE, surveyed homeowners adjacent to trails. Nearly two-thirds of respondents who purchased their home after the trail was built said that the trail positively influenced their purchase decision. Eighty-one percent felt that the nearby trail's presence would have a positive effect or no effect on the sale of their homes (Greer 2000). A 2008 study found that the Little Miami Scenic Trail (in Ohio) positively impacts single-family residential property values, with sale prices increasing by $7.05 for every foot closer a property is located to the trail (Karadeniz 2008).

Walkability is also an asset for homeowners.

> *Reduced traffic noise, traffic speeds, and vehicle-generated air pollution can increase property values. One study found that a 5- to 10-mph reduction in traffic speeds increased adjacent residential property values by roughly 20%. Another study found that traffic restraints that reduced volumes on residential streets by several hundred cars per day increased home values by an average of 18% (Local Government Commission 2000).*



Savings

According to the Bureau of Labor Statistics, in 2009 transportation was second only to shelter for household expenditures. The average American household spent 16 cents of every dollar on transportation.

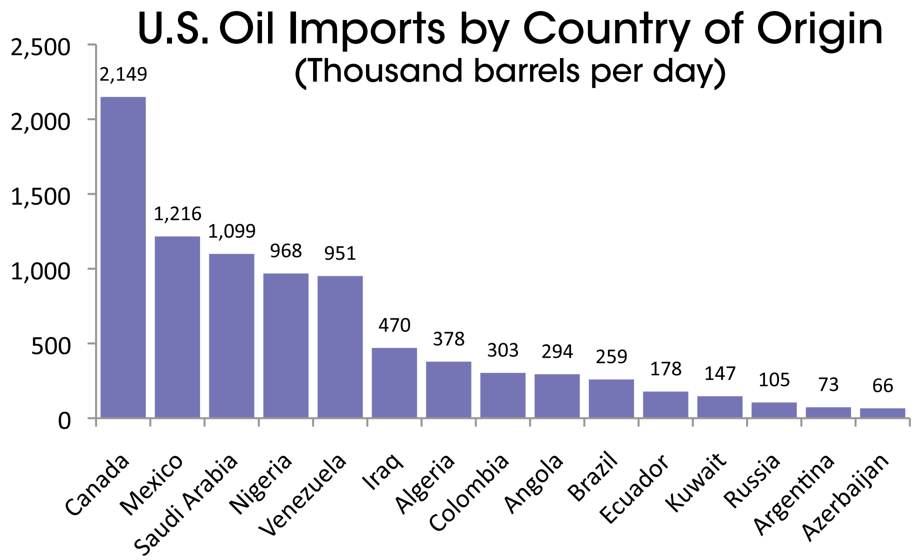
There is no denying that walking and bicycling are two of the least expensive ways to travel. In 2011, the American Automobile Association (AAA) estimated the average cost of owning and operating a car at $7,632 a year for a person driving 10,000 miles/year and paying $2.88 per gallon of gas (AAA 2011). In May 2011 gas had already reached over $4 per gallon in several states.

Much of this money goes to foreign car companies, and to pay for foreign fuel. According to the U.S. Energy Information Administration, 66% of the cost of gasoline is the crude oil. In January 2011 the United States imported over 370 million barrels of oil from a number of countries. The chart above shows the

Nonmotorized Transport Is Generally Cheaper Than Alternatives

Affordable and Efficient	Expensive and Resource Intensive
Walk and bike for transport	Own and operate an automobile
Walk and bike for exercise	Join a health club
Walk and bike children to school	Chauffeur children to school
Build sidewalks	Build roads and parking facilities

Source: Table above reproduced with permission from Todd Litman, Economic Value of Walkability, Victoria Transport Policy Institute, December 2007



Source: U.S. Energy Information Administration, Crude Oil Imports Top 15 Countries, April 2011.

countries whose economies are benefitting the more Americans drive.

Building infrastructure for bicycling and walking is also affordable. For the cost of 1 mile of four-lane urban highway, hundreds of miles of pedestrian and bicyclist facilities can be built. This investment, approximately $50 million, could complete the active transportation network of a mid-sized city (Gotschi 2008).

Bay Area Household Transportation Expenditures
(Adjusted for inflation to 2011 Dollars)

Mode Choice	Annual Spending	Savings Over Driving
Auto ownership *Vehicle purchases* *Other vehicle expenses* *Gasoline and motor oil*	$9,054 (breakdown below) *$3,632* *$3,892* *$1,530*	N/A
City Car Share	$1,090 ($91/month)	$7,964
MUNI (bus) monthly Fast Pass	$623 ($52/month)	$8,431
BART (train)	$545 ($45/month)	$8,509
Bike Riding	$26-390	$8,664-$9,028
Walking	$0	$9,054

Source: Table above reproduced with permission from Emily Drennan, 2003 **Note:** Data used for original table was from 2000, data in this table has been adjusted for inflation to 2011 dollars. This does not necessarily represent the current cost of each mode of transport as some pricing may have risen faster or slower than inflation.

Communities that invest in bicycling and walking are investing in their local economies. Bicycling and walking for transport results in reduced transportation expenses. A 2003 report examined the annual spending of various modes of transport in the San Francisco Bay Area and the savings of each mode over driving. Bicyclists and pedestrians save the most—between $8,664 and $9,054 a year (adjusted to 2011 dollars). This equals more disposable income to invest locally in goods, services, and entertainment (Drennan 2003).

Places where residents can easily choose to bike, walk, or take transit may have more thriving local economies. A 2010 study looked at the impact of walkability on New York City's economy: *"According to the Internal Revenue Service data, about 73 percent of the retail price of gas (back when it was under $2.00 a gallon, by the way) and 86 percent of the retail price of cars is the 'cost of goods sold,' which immediately leaves the local economy. The $19 billion New Yorkers save on car travel translate into* *more than $16 billion that are available to be spent in the local economy. Because this money tends to be re-spent in other sectors of the economy, it stimulates businesses"* (Cortright 2010).

Reducing car dependence is investing in local business.

In addition, pedestrian improvements encourage residents to shop locally in their own neighborhoods. One successful example is a $4.5 million public-private pedestrian-oriented project in Lodi, CA. The city credits the retrofit of five main street blocks with widened sidewalks, bulbed-out intersections, and other improvements for a large economic turnaround. Vacancy rates dropped from 18 to 6 percent. And, after the work was completed, the city saw a 30% increase in downtown sales tax revenue (Local Government Commission 2000).

Cost / Benefit

Besides the obvious impact on job creation, tourism, and savings, there are other economic benefits of bicycling and walking that are harder to quantify. Some of these include reduced traffic congestion, reduced time spent in traffic, improved health and reduced mortality, cleaner air, reduced noise pollution, and lower levels of stress.

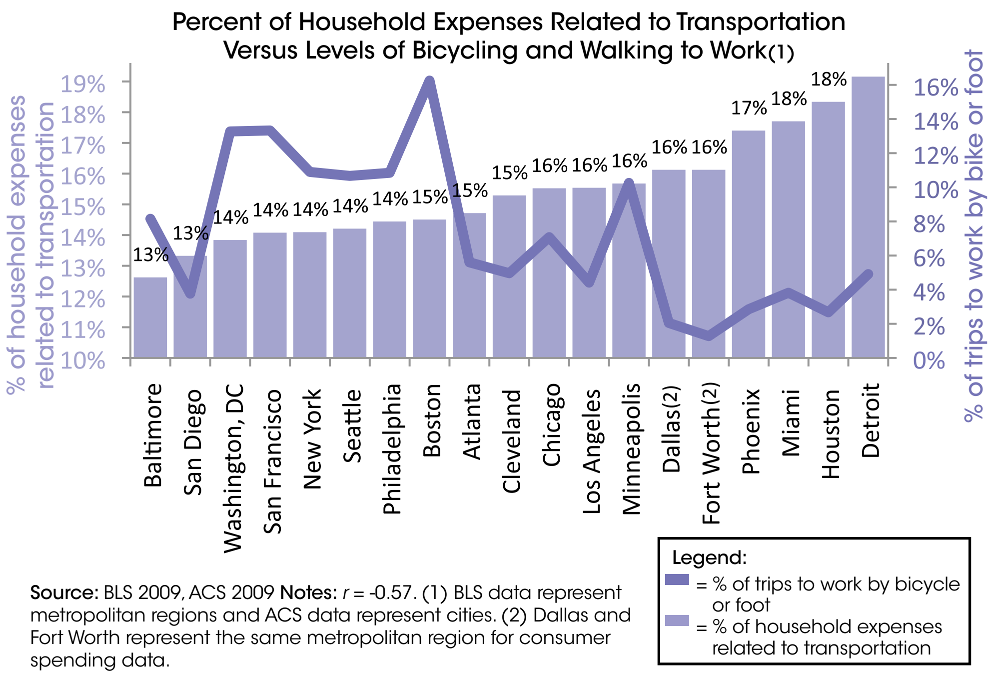
A 2008 report by Rails-to-Trails Conservancy attempted to quantify, for the first time, the benefits the United States could expect for elevating the priority of bicycling and walking in our transportation policy. Benefits were quantified for the status quo and for increases in bicycling and walking under a "modest" and "substantial" scenario. The results of increasing active transportation were significant. The modest scenario showed a reduction of 70 billion miles of automobile travel while the substantial increase in bicycling and walking could help divert 200 billion miles per year.

Valuing bicycling and walking's impact on health

Bicycling and walking for transportation has a great potential to influence public health. Because bicycling and walking are part of a healthy active lifestyle, these activities have been shown to exist more alongside reduced levels of obesity. Further, because most American adults gain weight gradually—about 2 pounds each year—bicycling or walking for less than 30 minutes daily can ward off this extra weight (Gotschi 2008).

According to a 2010 study sponsored by the Society of Actuaries, the total economic cost of overweight and obese citizens in the Unites States and Canada was roughly $300 billion in 2009. This estimate includes medical costs, disability, and excess mortality (Behan et al., 2010).

A 2006 report by the National Governors Association found that obesity costs the average taxpayer $180 per year regardless of their own health status. Studies show that promoting physical activity is cost-effective and the value of health benefits can far outweigh the costs (Gotschi 2008, Gotschi 2011, National Governors Association 2006, Roux et al., 2008). Further, if just one of every ten adults started a regular walking program, the United States could save $5.6 billion —the equivalent of paying the college tuition of 1,020,000 students (National Governors Association 2006). (To see how your state compares, see the table to the right.)



Source: BLS 2009, ACS 2009 **Notes:** *r* = -0.57. (1) BLS data represent metropolitan regions and ACS data represent cities. (2) Dallas and Fort Worth represent the same metropolitan region for consumer spending data.

Annual Household Transportation Expenses

City (1)	Annual household transportation expenditures	Vehicle purchases	Gasoline and motor oil	Other vehicle expenses	Public transportation	Percent of expenditures related to transportation
Atlanta	$6,760	$1,597	$2,631	$2,171	$360	15%
Baltimore	$6,621	$1,452	$2,444	$2,178	$547	13%
Boston	$8,591	$2,818	$2,125	$2,913	$735	15%
Chicago	$8,840	$3,101	$2,364	$2,636	$739	16%
Cleveland	$7,010	$2,098	$2,049	$2,409	$454	15%
Dallas (2)	$8,689	$2,877	$2,616	$2,806	$390	16%
Detroit	$9,463	$2,793	$2,624	$3,674	$373	19%
Fort Worth (2)	$8,689	$2,877	$2,616	$2,806	$390	16%
Houston	$10,843	$3,874	$2,980	$3,421	$568	18%
Los Angeles	$8,784	$2,513	$2,667	$2,989	$615	16%
Miami	$8,427	$2,921	$2,680	$2,318	$508	18%
Minneapolis	$8,833	$2,911	$2,350	$2,973	$598	16%
New York	$8,495	$2,321	$1,943	$3,137	$1,093	14%
Philadelphia	$8,202	$2,037	$2,240	$3,156	$769	14%
Phoenix	$9,330	$2,887	$2,658	$3,275	$509	17%
San Diego	$7,171	$1,941	$2,412	$2,373	$445	13%
San Francisco	$9,535	$2,748	$2,235	$3,252	$1,300	14%
Seattle	$9,380	$3,395	$2,454	$2,589	$943	14%
Washington, DC	$9,563	$3,028	$2,465	$2,864	$1,206	14%
Mean/Average (3)	$8,591	$2,680	$2,450	$2,839	$660	15%
Median	$8,689	$2,818	$2,454	$2,864	$568	15%
High	**$10,843**	**$3,874**	**$2,980**	**$3,674**	**$1,300**	**19%**
Low	**$6,621**	**$1,452**	**$1,943**	**$2,171**	**$360**	**13%**

Legend:
= High value
= Low value

Source: BLS 2009 **Notes:** (1) BLS data are for metropolitan regions. (2) Dallas and Fort Worth represent the same metropolitan region for consumer spending data. (3) Average here represents the average among the cities listed and not the national average.

The 2008 report by Rails-to-Trails Conservancy estimated the health-related cost savings of a modest increase in bicycling and walking to be $420 million annually. A substantial increase was estimated to save over $28 billion per year. This includes reduced health costs from an increase in physical activity by those who currently do not meet recommended levels (Gotschi 2008).

A 2011 study found that Portland, OR would see between $388 and $594 million in health cost savings attributable to new bicycle infrastructure and programs by 2040. The savings in value of statistical lives was between $7 and $12 billion (Gotschi 2011).

Valuing bicycling and walking's impact on air quality and greenhouse gases

Reducing vehicle miles traveled also adds up to cleaner air. Communities designed to encourage safe bicycling and walking help reduce driving and thereby reduce fuel consumption and air pollution associated with automobiles. This amounts to reduced smog that contributes to respiratory illness and asthma and reduced greenhouse gases that contribute to global warming. Although these savings and reductions are hard to quantify monetarily, some studies have attempted estimates.

The 2008 study by Rails-to-Trails Conservancy estimated that a modest increase in bicycling and walking would save 3 billion gallons of gasoline and keep 28 million tons of CO_2 from the atmosphere. A substantial increase could save 8 billion gallons of gas and avoid 73 million tons of CO_2. According to the report:

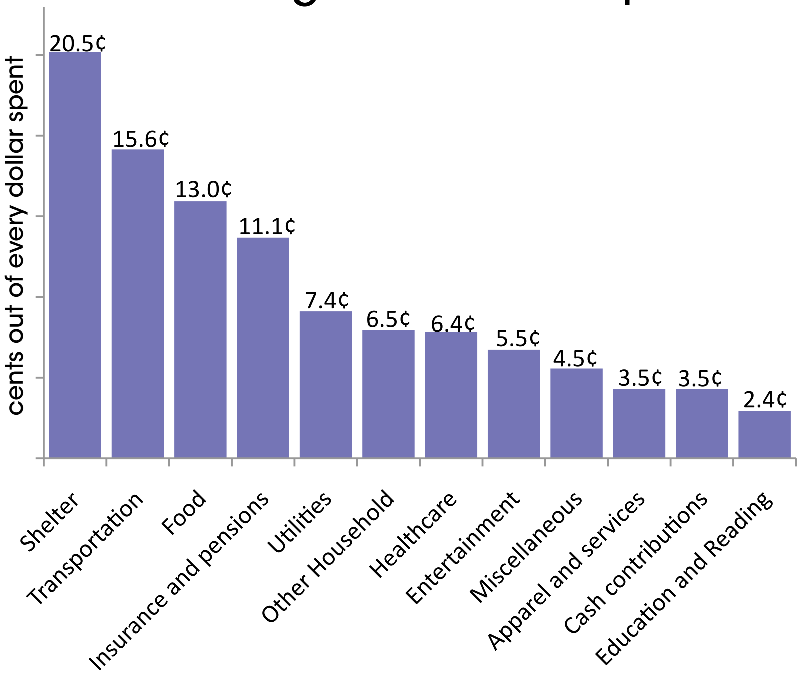
(Continued page 192)

Obesity Costs and Potential Health Savings

State	Obesity-related costs per taxpayer per year	State savings (millions) if 1 in 10 adults started walking program	Number of students for which college tuition could be paid
Alabama	$195	$79	14,387
Alaska	$214	$19	3,460
Arizona	$89	$86	15,662
Arkansas	$161	$48	8,742
California	$147	$604	109,998
Colorado	$128	$52	9,470
Connecticut	$162	$82	14,934
Delaware	$160	$21	3,824
Florida	$148	$241	43,890
Georgia	$164	$168	30,596
Hawaii	$152	$33	6,010
Idaho	$114	$18	3,278
Illinois	$183	$208	37,880
Indiana	$177	$86	15,662
Iowa	$177	$43	7,831
Kansas	$163	$43	7,831
Kentucky	$186	$79	14,387
Louisiana	$209	$99	18,030
Maine	$177	$33	6,010
Maryland	$185	$108	19,669
Massachusetts	$186	$121	22,036
Michigan	$196	$181	32,963
Minnesota	$174	$109	19,851
Mississippi	$179	$67	12,202
Missouri	$191	$121	22,036
Montana	$127	$15	2,732
Nebraska	$176	$34	6,192
Nevada	$97	$26	4,735
New Hampshire	$155	$22	4,007
New Jersey	$179	$199	36,241
New Mexico	$118	$39	7,103
New York	$210	$698	127,117
North Carolina	$166	$166	30,231
North Dakota	$220	$12	2,185
Ohio	$193	$209	38,062
Oklahoma	$164	$54	9,834
Oregon	$144	$60	10,927
Pennsylvania	$219	$295	53,724
Rhode Island	$185	$29	5,281
South Carolina	$169	$86	15,662
South Dakota	$173	$12	2,185
Tennessee	$207	$126	22,947
Texas	$165	$396	72,118
Utah	$121	$24	4,371
Vermont	$150	$13	2,368
Virginia	$146	$85	15,480
Washington	$144	$121	22,036
West Virginia	$208	$35	6,374
Wisconsin	$181	$84	15,298
Wyoming	$116	$11	2,003
U.S. Total	$180 (1)	$5,600	1,020,000

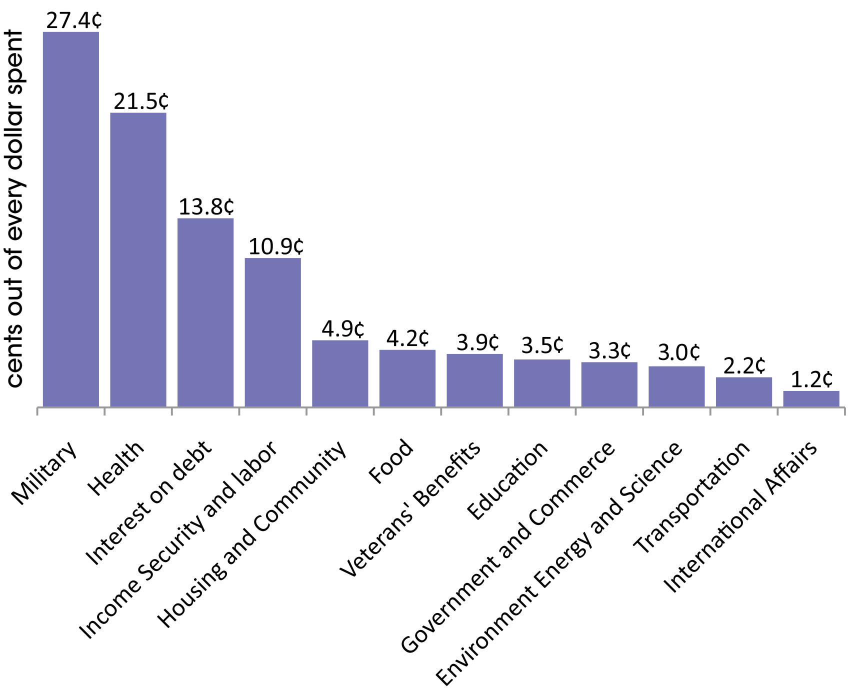
Source: National Governors Association 2006. http://www.nga.org/Files/pdf/0608HEALTHYREPORTNH.PDF **Note:** (1) Average per U.S. taxpayer.

How an Average Household Spends a Dollar



Source: BLS 2009 **Notes:** Shelter includes mortgages, taxes, maintenance,home insurance, and rent; Other Household includes housekeeping supplies, household furnishings, and equipment; Miscellaneous includes personal care products and services, alcohol, tobacco products, and other miscellaneous expenditures.

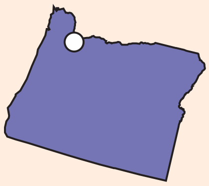
How the Federal Government Spends a Tax Dollar



Source: National Priorities Project 2011 **Notes:** The breakdown of the federal income tax dollar is based on an analysis of each agency's Federal Fund outlays according to function and subfunction for FY2010. Figures are from Budget of the United States Government, FY2012, Public Budget Database.

CLOSER LOOK

Cost/Benefits of Bicycling Investment in Portland, OR



Increasing bicycling is widely recognized as a worthy objective with many benefits including public health. However, just how much benefit comes from investing in bicycling is largely unknown. A 2011 study by Thomas Gotschi published in the *Journal of Physical Activity and Health* sought to quantify the benefits of investing in bicycling using Portland, OR, as a case study.

Portland, OR, has the highest levels of bike commuting to work of any major U.S. city and has seen the greatest increases in bike to work mode share of any major U.S. city over the last decade. In 2008, Portland estimated the cost of rebuilding its 274-mile bicycle network at $57 million. According to Mayor Sam Adams, "In 1993 we weren't the bicycling capital of America. Seventeen years later, for the equivalent cost of a single mile of freeway, we have a bike infrastructure."

Gotschi took Portland's estimated replacement cost of their bike network and added the cost of the Smart Trips program which encourages bicycling, walking, and transit use (another $7.2 million through 2012). He then modeled growth in bicycling and associated benefits based on three scenarios over a 50-year period: a status quo "basic" $100 million plan proposed in the context of the renewal of the federal transportation bill, and two plans included in the recently adopted 2030 Portland Bike Master Plan. These include the "80% plan" with over $329 million invested to put 80% of residents within a quarter mile of a low-stress bikeway, and a $773 million “world class” plan.

Among others, these plans foresee investments in crucial trail sections, bicycle boulevards (traffic calmed streets which limit motorized through traffic and specifically accommodate bicycles), cycle tracks (bicycle lanes physically separated from traffic), bicycle and

Bicyclists ridie Portland's annual Bridge Pedal.
Photo by Thomas Le Ngo

For the cost of a single mile of freeway, Portland built an entire bicycle network.

Bike box in Portland, OR. Photo by Arthur Wendall.

pedestrian bridges, various improvements and maintenance of existing infrastructure, the continuation of Smart Trips, and several additional projects.

Gotschi concluded that even when considering just health care cost savings and fuel savings, the investments in bicycling yield benefits between 3.8 and 1.3 times the cost. In addition, under the first two scenarios, the costs would be recouped by 2015 and by 2032 with the "world class" plan. Gotschi acknowledges the benefits included in his analysis do not give the full picture:

Accounting for lives saved from a reduction in mortality using value of statistical life, as is commonly done for transportation projects, dramatically increases the benefits-cost ratio. Including additional, less easily monetizable benefits would further bolster the economic case for investments in bicycling.

The study did not include the benefits of Portland's investments in cycling on other aspects of its economy. A 2006 survey asked over 100 Portland businesses if Portland's reputation for bike-friendliness is good for business. Eighty-two percent responded "Absolutely/yes" and only 3% responded "no." Another report by Alta Planning and Design found the direct impact of bicycle-related economic activity was $90 million in 2008, a 38% increase over 2006. In addition, the study found bicycle rides, events, and tours had nearly doubled since 2006—from 2,100 to nearly 4,000 annually.

These studies suggest that Portland's investment in bicycling will pay for itself many times over in reduced public health and fuel costs, and increased economic activity.

Sources: Alta Planning and Design 2008, Gotschi 2011, Maus 2006

Looking Outside the Borders

Bicycling in Groningen: An Economic Program

This article was originally published by Global Ideas Bank and is reprinted here with permission from Andrew Curry, Global Ideas Bank, April 2004.

In Groningen, the Netherlands' sixth largest city, the main form of transport is the bicycle. Sixteen years ago, ruinous traffic congestion led city planners to dig up city-center motorways. In 2003 they set about creating a car-free city center. Now Groningen, with a population of 170,000, has the highest level of bicycle usage in the West. Fifty-seven percent of its inhabitants travel by bicycle—compared with four percent in the UK.

" This is not an environmental program, it is an economic program. We are boosting jobs and business."

The economic repercussions of the program repay some examination. Since 1977, when a six-lane motorway intersection in the city's center was replaced by greenery, pedestrianisation, cycleways, and bus lanes, the city has staged a remarkable recovery. Rents are among the highest in the Netherlands, the outflow of population has been reversed and businesses, once in revolt against car restraint, are clamouring for more of it. As Gerrit van Werven, a senior city planner, puts it, 'This is not an environmental program, it is an economic program. We are boosting jobs and business. It has been proved that planning for the bicycle is cheaper than planning for the car.' Proving the point, requests now regularly arrive from shopkeepers in streets where 'cyclisation' is not yet in force to ban car traffic on their roads.

A vital threshold has been crossed. Through sheer weight of numbers, the bicycle lays down the rules, slowing down traffic, determining the attitudes of drivers. All across the city roads are being narrowed or closed to traffic, cycleways are being constructed and new houses built to which the only direct access is by bicycle. Out-of-town shopping centres are banned. The aim is not to force cars to take longer detours, but to provide a 'fine mesh' network for bicycles, giving them easy access to the city center.


Cyclists in Groningen.
Photo by Jan Glas.

Cyclists in Groningen.
Photo by Tup Wanders

Like the Netherlands nationally, Groningen is backing bicycles because of fears about car growth. Its ten-year bicycle programme is costing £20m, but every commuter car it keeps off the road saves at least £170 a year in hidden costs such as noise, pollution, parking and health.

Bicycling in Groningen is viewed as part of an integral urban renewal, planning and transport strategy. Bicycle-friendly devices seen as exceptional in the UK—separate cycle ways, advanced stop lines at traffic lights, and official sanction for bicyclists to do right hand turns at red lights—are routine.

New city center buildings must provide bicycle garages. There are tens of thousands of parking spaces for bicycles, either in 'guarded' parks—the central railway station has room for over 3,000—or street racks. Under the city hall a nuclear shelter has been turned into a bicycle park (bike parking).

"We don't want a good system for bicycles, we want a perfect system", says Mr. van Werven. "We want a system for bicycles that is like the German autobahns for cars. We don't ride bicycles because we are poor—people here are richer than in England. We ride them because it is fun, it is faster, it is convenient."

To achieve equivalent fuel savings through vehicle efficiency improvements alone, between 19 million and more than 50 million drivers would need to trade in their vehicles for a highly efficient gas-electric hybrid version of the same model. To put this in perspective, there are about 250 million automobiles on America's streets and, despite rapidly growing sales, by the end of 2007 only about one million of them were hybrid vehicles."

The study estimated the value of CO_2 reduction from miles driven avoided to be $333 million with a modest increase in bicycling and walking and over $2.7 billion with a substantial increase (Gotschi 2008).

A Prudent Investment

Research looking at the cost/benefit of investing in bicycling and walking all point to the same conclusion. Bicycling and walking pay dividends.

- Lincoln, Nebraska: Every $1 spent on use of bicycle and pedestrian trails (including construction, maintenance, equipment, and travel) yields $2.94 in direct medical benefits (Wang et. al., 2008).
- Portland, Oregon: Every $1 invested in bicycling yields $3.40 in health care cost savings. When the statistical value of lives is considered, as is done for the evaluation of highway safety improvement projects, every $1 invested yields nearly $100 in benefits (Gotschi 2011).
- Kansas City: Every dollar invested in bicycle and pedestrian projects yields $11.80 in benefits (Ridgway 2010).
- Every dollar invested in bicycle networks yields at least $4 to $5 in benefits (Sælensminde 2004).

Investing in bicycle and pedestrian infrastructure is not only affordable, it creates jobs, stimulates local economies, increases property values, and has lasting benefits to health, air quality, and quality of life.


Photo by Elizabeth Edwards

Photo courtesy of the Safe Routes to School National Partnership

10: CONCLUSION

This report shows that increasing bicycling and walking are goals that are clearly in the public interest. Where bicycling and walking levels are higher, obesity, high blood pressure, and diabetes levels are lower. Higher levels of bicycling and walking also coincide with increased bicycle and pedestrian safety and higher levels of physical activity. Increasing bicycling and walking can help solve many serious problems facing our nation. As this report indicates, many states and cities are making progress toward promoting safe access for bicyclists and pedestrians, but much more remains to be done.

This report has highlighted numerous measures to promote bicycling and walking. As Chapter 7 discusses, a variety of policy measures and provisions are likely needed to make communities more bicycle and pedestrian friendly. Just as it took a large investment of public money into roads, signals, signs, and education for motorists, so too will it take an ongoing commitment of public investment in bicycling and walking to see major shifts toward these modes.

It is also crucial that the United States look to other countries to see what mode share levels are possible, and how

Photo by Paul Adkins

they have increased bicycling, walking, and safety. The United States lags far behind other countries and international cities in regard to walk and bike share of trips, safety, and public health. As this report has shown, the countries and cities with the greatest levels of bicycling and walking are also the safest places to bicycle and walk. These countries also have the lowest levels of obesity and report that prioritizing bicycling and walking is good for their economies.

As this report shows, the United States overall has great disparities between bicycling and walking mode share, safety, and funding. Twelve percent of trips are by bicycle or foot, yet bicyclists and pedestrians make up 14% of traffic fatalities and receive just 1.6% of federal transportation dollars. An international comparison of bicycle funding and mode share by Gotschi and Mills and Rails-to-Trails Conservancy (see chapter 4, page 96) demonstrates that international cities that invest greater amounts per capita into bicycling have greater levels of bicycling. These cities provide strong evidence that in order to increase bicycling and walking, the United States must invest significantly more in these modes.

Although greater investment in bicycling and walking is the primary recommendation of this report, there are many other measures that must be taken to simultaneously strengthen public policy, infrastructure, and behavior toward bicycling and walking. Over one-third of the U.S. population is under age 16 (typically cannot legally drive) or over age 65. Streets that do not adequately accommodate bicyclists and pedestrians create barriers for people who do not drive. This limits their ability to fully participate in American society, or else makes them reliant on others to drive them around. Less than half of states and major U.S. cities have adopted complete streets policies, which require that roadways be designed and built with all users in mind. In the absence of a national complete streets policy, the Alliance encourages states and jurisdictions to pursue local policies to begin to transform their local transportation culture and guarantee access for all road users.

Other policies featured in this report, such as education for police officers and the inclusion of bicycling and walking safety in driver education, are also key to the shift toward a bicycle and pedestrian friendly culture. Adult and youth education programs, public awareness campaigns such as "Share the Road," and other promotional efforts can also help raise awareness and change attitudes around bicycling and walking.

Many of the benchmarks featured in this report contribute to making com-

12% of trips in the United States are by bicycle or foot, yet bicyclists and pedestrians make up 14% of traffic fatalities and receive just 1.6% of federal transportation dollars.

munities more bicycle and pedestrian friendly by changing the built environment, culture, attitudes, and behaviors. But continuous evaluation of efforts to promote bicycling and walking is key to better understanding the relationships between levels of bicycling and walking, safety, policies, provisions, advocacy capacity, and other measures. Benchmarking is a necessary process to better understand these relationships, identify the most strategic areas on which to focus resources, and ultimately to increase these forms of active transportation.

Through researching and compiling data for this report, the Alliance has identified several places in need of improved data collection.

- Accurate data on state and local bicycling and walking levels are the key outcome indicator needed to measure success, and yet none currently exists. The ACS is an annual survey that measures share of commuters by bicycle or on foot, but no existing survey collects data on all trip purposes with enough samples to evaluate the local level. A national household travel survey should be funded to allow for greater sample size at the local level and to be conducted more regularly—every 7 years is not frequent enough. The Alliance encourages the federal government to fund annual or biennial travel surveys to more accurately measure progress and to better understand travel behaviors, trends, and needs for all modes of transportation.

- Funding data on bicycling and walking are recorded by state departments of transportation for the FHWA. Differences in project coding methods mean that data on funding are not always accurate or comparable between states. For example, a project with bicycle and pedestrian components that are a small amount of the total project cost might not be coded as a bicycle and pedestrian project at all in one location, but another location might break the project out into its parts. Also, some projects are coded by county, some by standard place code, and some by urbanized area. The differences in coding make it difficult to identify which projects are in certain cities. If projects that spanned a county also included codes for the cities affected by the project, it would be easier to obtain accurate spending data at the local level. The Alliance encourages the FHWA to set standards for coding projects so that spending on bicycling and walking can be more accurately tracked.

- The FHWA should develop a better method of tracking federal safety funding and what percentage of this funding in each state is being used for bicycle and pedestrian projects. With the great disparities that exist between bicycle and pedestrian mode share and fatality rates, it is essential that officials and advocates push for a fair share for safety.

- Many states and cities were unable to provide data on quantities of bicycle and pedestrian infrastructure such as miles of sidewalks, bike lanes, trails, and number of bicycle racks. The Alliance recommends that the FHWA develop a framework for best practices for states and local jurisdictions to conduct audits and report on bicycle and pedestrian facilities (and gaps) every 1 to 2 years.

Crosswalk at Shibuya, Tokyo, Japan
Photo by dog company @ Flickr.

- States and cities should produce a document every 1 to 2 years indicating the shortfall in funding needed to complete their bicycle and pedestrian system. This would provide vital data on cost needs, something that has existed for highways and bridges, but not for bicycle and pedestrian facilities.

- Tracking of participation levels in education and encouragement events is sparse. Evaluation is a key component to measuring the success or impact of these efforts. The Alliance encourages all states and cities to track participation levels and other outcomes associated with these encouragement and education programs. For example, a city could report on how many people participated in Bike to Work Day and through a survey sample could ask what influence this event had on their biking to work on this day and in the future. These sorts of measurements, tracked over time, could help evaluate the program's effectiveness.

In the future, the Alliance hopes to expand the Benchmarking Report to include other measures affected by bicycling and walking such as reductions in greenhouse gas emissions. The Alliance also hopes to include other international data for comparison purposes, to help demonstrate the great strides that U.S. states and cities must take to meet the ambitious goals they are setting for themselves (for increasing bicycling and walking mode share).

In the meantime, this report provides plenty of examples of states and cities that are leaders in a variety of efforts to promote bicycling and walking. Appendix 5, page 209, lists a number of resources from these cities and states. These are presented so that practitioners can have models for inspiration when working toward their goals.

The Benchmarking Report should be used as a tool by cities and states to learn what works best to promote bicycling and walking and what is possible here in the United States. States and cities can learn from each other's successes and failures and set their goals accordingly. The Alliance encourages all state and city officials and advocates to take an active role in benchmarking their efforts to promote bicycling and walking. Even smaller cities that are not included in this report can collect data from their city and compare it to the benchmarks and progress in other communities.

There is no doubt that government officials and advocates seeking to grow bicycling and walking have a lot of work ahead of them. But it is crucial for advocates and officials to take the time to evaluate their efforts. While many international benchmarking efforts require huge investments of government time and money to participate, the Alliance's Benchmarking Project is a free service that requires a relatively small amount of time to complete a survey every 2 years. With more officials and advocates taking the time to fully participate, this project will become a better source of information and a stronger benchmarking tool for everyone.

If you would like more information about this report, please contact the Alliance at benchmarking@PeoplePoweredMovement.org.

Photo courtesy of Minnesota Clean Energy Resource Teams

Appendix 1: Overview of Data Sources

Data Source	Description	Method of Data Collection	Frequency of Data Collection	Last Date Available(1)
ACS	American Community Survey: a survey conducted by the U.S. Census Bureau that annually collects year-round data	Similar to Census long form; (about three million households)	Continuous	2009
APTA	American Public Transportation Association—Public Transportation Vehicle Database: collects and summarizes data on transit agency vehicles	Data are from the National Transit Database (NTD) report published by the U.S. Federal Transit Administration (FTA). APTA supplements these data with special surveys.	Yearly	2010
BRFSS	Behavioral Risk Factor Surveillance System: from Centers for Disease Control and Prevention (CDC); statewide health information	Telephone health survey	Continuous	2009
Census	From U.S. Census Bureau	Mailed forms, and house visit for nonresponders	Every 10 years	2010
FARS	Fatality Analysis Reporting System: federal database of the National Highway Traffic Safety Administration (NHTSA) of vehicle injuries and fatalities	FARS analyst from each state collects data from governments	Yearly	2009
FHWA - FMIS	Federal Highway Administration (FHWA) Fiscal Management Information System (FMIS)	Data reported to FHWA from state and local government agencies	Continuous	2010
LAB	League of American Bicyclists: Bicycle Friendly State program surveys collect information on statewide policies, education, enforcement, and other efforts aimed at bicycle promotion	Online surveys sent to state bicycle and pedestrian coordinators	Yearly	2011
NCSRTS	National Center for Safe Routes to School: (Walk To School Day Participation) tracks numbers of schools signed up to participate (Safe Routes to School [SRTS] National Program): Quarterly SRTS Program Tracking Brief provides information about state SRTS programs	(Walk to School Day): online form completed by event organizer (SRTS National Program): questionnaires to state Safe Routes to School Coordinators	(Walk to School Day): Continuous (SRTS National Program): Quarterly	2011
NCSC	National Complete Streets Coalition: tracks and assists with complete streets policies	Monitors adoption of policies through network, media, etc.	Continuous	2011
NHANES	National Health and Nutrition Examination Survey: studies designed to assess the health and nutritional status of adults and children in the U.S.; program of the National Center for Health Statistics (NCHS) and the CDC	Interviews and physician examinations	Continuous	2005-2006
NHIS	National Health Interview Survey: estimates of broad range of health measures	Interviews at households	Yearly	2005
NHTS	National Household Travel Survey: inventory of daily and long-distance travel; NHTS is a national survey, and analysis below the national level have problems with small samples; also, NHTS data is reported by metropolitan areas so data shown for cities are estimates only	Survey of 26,000 households (additional 44,000 from nine "add-on" areas); collected by the FHWA	Every 5-7 years since 1969	2009
NTEC	National Transportation Enhancements Clearinghouse: sponsored by the FHWA and Rails-to-Trails Conservancy, reports on funded projects	Information comes from funded Transportation Enhancement (TE) projects	Yearly	2010
RTC	Rails-to-Trails Conservancy: tracks current information about the trails movement and rail-trail use at the national and state level	Monitors rail trails through media, interviews with trail managers, and network	"Periodically"	2/2011
SRTSNP	Safe Routes to School National Partnership: monitors and collects benchmarking data on the national Safe Routes to School program and produces quarterly "State of the States" report	Secondary data collection: from the Federal Highway Administration and other sources	Quarterly	2011
STN	School Transportation News: inventory of U.S. transportation data elements on a state-by-state basis, specifically including student enrollment and school bus information	Surveys to the pupil transportation section of state departments of education	Yearly	2011
USHCN	United States Historical Climatology Network: daily and monthly meteorological data	1,000 observing stations	Continuous	2004-2005

(1) Latest date of availability, presented in this report, as of June 2011.

Appendix 2: Organization and Study Area Matches

City Advocacy Organizations

City	Alliance organization
Albuquerque	Bike ABQ (1)
Arlington, TX	NRO
Atlanta	Atlanta Bicycle Coalition
Austin	League of Bicycling Voters, Austin Cycling Association(1)
Baltimore	NRO
Boston	1- LivableStreets Alliance, 2- Walk Boston★ 3- Boston Cyclists Union★
Charlotte	Charlotte Area Bicycle Alliance
Chicago	1- Active Transportation Alliance, 2- Chicagoland Bicycling Group★
Cleveland	1- Walk+Roll, 2- Cleveland Bikes
Colorado Springs	NRO (4)
Columbus	1- Consider Biking, 2 -Yay Bikes★
Dallas	BikeDFW
Denver	BikeDenver
Detroit	NRO
El Paso	NRO
Fort Worth	BikeDFW
Fresno	NRO
Honolulu	Hawaii Bicycling League
Houston	Citizens' Transportation Coalition★
Indianapolis	1- Alliance for Health Promotion★ 2- IndyCog
Jacksonville	NRO
Kansas City, MO	Kansas City Bicycle Club, Bike Walk KC (1)★, Revolve (1)★
Las Vegas	Outside Las Vegas Foundation(1)★
Long Beach	Los Angeles County Bicycle Coalition, City Fabrick (1)★, Bikeable Communities (1)★
Los Angeles	Los Angeles County Bicycle Coalition, C.I.C.L.E. (1), Bikeside(1)
Louisville	Bicycling for Louisville
Memphis	Livable Memphis★
Mesa	NRO
Miami	Green Mobility Network
Milwaukee	Bicycle Federation of Wisconsin (3)
Minneapolis	1- St. Paul Smart Trips, 2- Midtown Greenway Coalition★, Minneapolis Bicycle Coalition (1)★
Nashville	NRO (4)
New Orleans	Bike Easy
New York	Transportation Alternatives
Oakland	Walk Oakland Bike Oakland
Oklahoma City	NRO
Omaha	Bike Omaha (1)
Philadelphia	Bicycle Coalition of Greater Philadelphia
Phoenix	NRO
Portland, OR	1- Bicycle Transportation Alliance (3), 2- Willamette Pedestrian Coalition★ 3- Community Cycling Center★
Raleigh	NRO
Sacramento	Sacramento Area Bicycle Advocates, Walk Sacramento
San Antonio	NRO
San Diego	San Diego County Bicycle Coalition
San Francisco	1- San Francisco Bicycle Coalition 2- Walk SF
San Jose	Silicon Valley Bicycle Coalition (1)
Seattle	1- Cascade Bicycle Club 2- Feet First 3- Undriving★
Tucson	Living Streets Alliance (1)★
Tulsa	Tulsa Hub★
Virginia Beach	NRO
Washington, DC	Washington Area Bicyclist Association (1)

Legend:
NRO = No Representative Organization ★= Organization is new to the Alliance since the 2010 Benchmarking Report

State Advocacy Organizations

State	Alliance organization
Alabama	Alabama Bicycle Coalition
Alaska	NRO
Arizona	Coalition of Arizona Bicyclists★
Arkansas	Bicycle Advocacy of Central Arkansas
California	California Bicycle Coalition
Colorado	Bicycle Colorado
Connecticut	Bike Walk Connecticut★
Delaware	Bike Delaware★
Florida	Florida Bicycle Association
Georgia	Georgia Bikes!
Hawaii	Hawaii Bicycle League
Idaho	Idaho Pedestrian and Bicycle Alliance★
Illinois	League of Illinois Bicyclists
Indiana	Bicycle Indiana
Iowa	Iowa Bicycle Coalition
Kansas	KanBikeWalk★ (1)
Kentucky	NRO (4)
Louisiana	NRO
Maine	Bicycle Coalition of Maine
Maryland	Bike Maryland
Massachusetts	MassBike
Michigan	League of Michigan Bicyclists
Minnesota	Bicycle Alliance of Minnesota, Parks and Trails Council of Minnesota (1)
Mississippi	Bike Walk Mississippi
Missouri	Missouri Bicycle and Pedestrian Coalition
Montana	NRO
Nebraska	NRO
Nevada	Nevada Bicycle Coalition★
New Hampshire	Bike-Walk Alliance of NH
New Jersey	New Jersey Bike+Walk Coalition
New Mexico	Bicycle Coalition of NM
New York	NY Bicycling Coalition
North Carolina	North Carolina Active Transportation Alliance★
North Dakota	NRO
Ohio	Ohio Bicycle Federation (1), Bike Walk Ohio (1)
Oklahoma	Oklahoma Bicycling Coalition
Oregon	Bicycle Transportation Alliance (3)
Pennsylvania	PA Walks and Bikes★
Rhode Island	Rhode Island Bicycle Coalition (1)
South Carolina	Palmetto Cycling Coalition
South Dakota	South Dakota Bicycle Coalition
Tennessee	Bike Walk Tennessee★
Texas	BikeTexas (1)
Utah	Bike Utah
Vermont	Vermont Bicycle and Pedestrian Coalition (1)
Virginia	BikeWalk Virginia (1)
Washington	Bicycle Alliance of Washington
West Virginia	NRO (4)
Wisconsin	Bicycle Federation of Wisconsin (3)
Wyoming	Teton Valley Trails and Pathways

Legend:
NRO = No Representative Organization
★ = Organization is new to the Alliance since the 2010 Benchmarking Report

Notes: These tables show 50 states and the 51 cities that were the study areas of this report. In Chapters 6 and 7 these organizations are cited by the state or city they represent for ease of comparison (and because not all organizations contain their city or state in their organization's name). NRO = No Representative Organization as of May 2011. (1) This organization did not provide data for this report and thus was not included in report illustrations and comparisons. (2) This state/city has formed a new Alliance advocacy organization since the time of data collection for this report. (3) This statewide organization also dedicates significant time and resources into a city and are representative of both the state and a city in this report. (4) This city or state formerly had a representative organization that has since dissolved or is no longer a member of the Alliance. States and cities with NRO do not have an Alliance member organization dedicated to bicycle and/or pedestrian advocacy in their area. This representation is by no means all inclusive. Only Alliance member organizations were surveyed for this report. Some areas are represented by more than one Alliance organization. Organizations that are not included in this report are not Alliance member organizations (state or local bicycle and pedestrian nonprofit organizations).

Appendix 3: Challenges with Trip Data

Determining How Many People Bike and Walk

The question of how many people bicycle or walk is not easily answered with the limited data available. The most reliable source of information on how many people bike or walk comes from the U.S. Census Journey to Work data (and the annual American Community Survey). However, census figures are limiting and inaccurate for a number of reasons. The Census Bureau only collects data on the main mode of transportation to work. This measure excludes trips of individuals not in the workforce, such as children or retirees. Moreover other trip purposes, such as shopping and recreational outings, are not captured. Additionally, the Census Bureau only reports the main mode of transportation to work, thus excluding many walk and bike trips used for shorter segments of commutes, for example, walk trips to transit stops or a walk from the parking garage to work, and also misses people who walk or bicycle 1 or 2 days a week.

Comparing Data from the Census and ACS Surveys

It is also not completely accurate to compare data from the decennial Census to the annual American Community Survey. While the decennial Census is taken in April, ACS data are collected throughout the year. The time of year the Census data are collected might influence reported bike and walk share of work trips. This is particularly true in cities such as Minneapolis and Boston which can still be cold in April. Although the decennial Census has a larger sample size, in this case, the ACS may more accurately reflect bicycle travel because it is collected throughout the year.

The biggest difference in the surveying between the ACS and the Census is that the ACS is done every year instead of every decade. However, the Census provides detailed socioeconomic data and for much smaller areas. There are differences in the ACS and the Census when it comes to residence rules, universes, and reference periods. However, comparisons can generally be made for most population and housing subjects. For some categories such as disability, income, and employment status, the U.S. Census Bureau recommends not comparing or comparing with caution. But according to the Bureau, the category "means of transportation to work" is comparable from the ACS to the Census and between the different years of the ACS. http://www.census.gov/acs/www/UseData/Comparison_Guidance.htm#transport.

Bicycle and Pedestrian Travel Data for All Trip Purposes

The National Household Travel Study (NHTS) is another source of data on daily travel, sponsored by the Bureau of Transportation Statistics and the Federal Highway Administration. The NHTS attempts to collect data on all trips, not just trips to work. However, because it is a national survey, analysis below the national level has problems with small sample sizes. It is also difficult to extract data for cities from this source as it uses Metropolitan Statistical Areas (MSAs), which often stretch beyond city boundaries. Also, the NHTS is only collected every 5 to 7 years. Due to these limitations, NHTS data on city and state levels should be considered as rough estimates for walking and bicycling in these areas.

The NHTS methodology includes a brief phone survey that gathers basic demographic information and asks the person if he or she is willing to keep a travel diary for a day to record all trips by members of the household, including children. Travel diaries are mailed to the household and NHTS officials follow up to answer any questions. Survey participants then receive a follow-up call from NHTS to collect information from the travel diary. They are asked a number of questions on their travel behavior during their assigned travel day and during the last week including such questions as how many times they went for a walk or bike ride, how long did they spend bicycling or walking, and (if they drive) how many minutes it takes them to walk from where they park to their workplace. To view the most recent complete NHTS questionnaire, visit http://nhts.ornl.gov/2008/doc/NHTS_2008_Questionnaire.pdf.

Other Trip Count Efforts

Because of the serious gap in reliable data on bicycling (and walking) trips, there have been numerous efforts to create a more reliable means to measure travel. Barnes and Krizek (2005) developed a formula for determining total bicycling trips by multiplying the commute share by 1.5 and adding 0.3%. Some cities have done their own travel counts in an attempt to determine the share of all bicycle trips. Of all cities surveyed, 25 reported having conducted some type of bicycle count at least once (Atlanta, Austin, Baltimore, Boston, Charlotte, Chicago, Columbus, Dallas, Denver, Honolulu, Houston, Long Beach, Louisville, Memphis, Minneapolis, New York, Oakland, Philadelphia, Portland, San Diego, San Francisco, San Jose, Seattle Tucson, and Washington, DC). Eight of these cities (Chicago, Columbus, Denver, Minneapolis, New York, Portland, San Francisco, and Seattle) reported having done an "all-trips" count to determine bicycle mode share.

San Francisco provides an example of the discrepancies in travel counts and methods to determine bicycling and walking mode share. The Barnes and Krizek formula indicates that 4.1% of all trips in San Francisco are by bicycle. This number is higher than the NHTS estimate of 0.93% of all trips represented by bicyclists. According to the 2000 Census, 1.98% of work trips are by bicycle. The Census Bureau's American Community Survey (ACS) 2007 data show bike to work share in San Francisco as 2.52%. And a city-commissioned study shows bicycling mode share is 6%. The study commissioned by San Francisco is more likely correct, because of a larger sample size and more robust methods. However, because there is a lack of standardized trip counts for multiple cities, the Alliance could not extrapolate a formula for all bicycle trips to apply across cities and states.

Prospective Solutions

The National Bicycle and Pedestrian Documentation Project (NBPD), coordinated by the Institute of Transportation Engineers, is attempting to address the gap in accurate and comparable trip count data. NBPD sets detailed standards and guidelines and provides tools for performing bicycle and pedestrian counts and surveys in communities and collects data from communities in a centralized database. By using the same methodologies, NBPD can compare progress of cities and better identify factors that influence bicycling and walking levels. To date, NBPD has collected counts from over 50 organizations and 500 locations. More widespread and consistent participation from local agencies could help this tool reach its full potential.

Another potential solution is for the federal government to fund more regular travel surveys. Currently the National Household Travel Survey is only done every several years and low sampling on the local level makes comparisons inaccurate. There is currently discussion in Congress about adding funding for improved data collection at the federal level as part of the next federal transportation bill. This sort of investment in surveying bicycling and walking levels is crucial to evaluating the impact of investments and efforts toward increasing these modes.

Applications

Collection of bicycling and walking data would assist transportation planners, public health officials, and elected officials in making informed decisions. Transportation planners would receive information regarding the impact of bicycling and walking facilities, and be able to put information on injuries in perspective with information on the levels of bicycling and walking. A robust data collection system could help public health officials target and assess community-level interventions for physical activity and injury prevention efforts. Elected officials would have access to the same types of data that exist for motor vehicles, including information on the cost of the projects and the subsequent effect on bicycling and walking.

The World Health Organization Regional Office for Europe has developed a promising tool, the Health Economic Assessment Tool (HEAT) for bicycling. This could further inform decisions about bicycling and walking infrastructure by providing an estimate for the economic value of positive health effects of bicycling. HEAT for bicycling requires information on the number of trips done by bicycle and the average trip distance, and, based on these inputs and best-evidence default values, the economic savings that result from reduced mortality due to regular physical activity from commuter bicycling are estimated. This tool could be used to estimate the value of health effects of current levels of bicycling, calculate the health-related economic benefits when planning new bicycling infrastructure, or provide input into more comprehensive cost-benefit analyses. If bicycling and walking data collection was as robust as other modes of transportation, it would assist professionals and the public to make better informed decisions about the design of their communities.

Appendix 4: Additional Data on Bicycling and Walking Commute Trends

Bicycle to Work Levels by City 1990-2009

City	# of daily bicycle commuters							% Change '90-'09	% Change '00-'09
	1990	2000	2005	2006	2007	2008	2009		
Albuquerque	2,174	2,408	1,918	2,857	1,878	4,384	3,596	65%	49%
Arlington, TX	234	294	433	137	256	644	75	-68%	-74%
Atlanta	487	562	955	1,108	1,367	1,106	2,732	461%	386%
Austin	1,885	3,280	4,654	3,468	3,833	5,545	4,458	136%	36%
Baltimore	761	824	1,018	552	824	1,620	2,654	249%	222%
Boston	2,456	2,705	2,377	3,495	2,900	4,946	7,156	191%	165%
Charlotte	335	417	481	118	133	822	537	60%	29%
Chicago	3,307	5,956	7,812	11,193	13,736	13,143	14,565	340%	145%
Cleveland	234	379	603	715	777	1,025	611	161%	61%
Colorado Springs	679	964	1,088	1,140	751	628	1,658	144%	72%
Columbus	1,212	1,242	2,131	1,803	2,598	3,361	2,661	120%	114%
Dallas	772	721	1,029	957	1,234	268	879	14%	22%
Denver	1,980	2,652	3,814	4,988	4,657	4,864	5,554	181%	109%
Detroit	340	507	547	928	812	796	1,217	258%	140%
El Paso	666	246	700	440	313	536	447	-33%	82%
Fort Worth	382	303	645	335	612	505	349	-9%	15%
Fresno	1,352	1,232	1,218	2,058	1,294	1,137	1,309	-3%	6%
Honolulu	2,376	2,155	2,504	2,690	1,833	2,944	4,565	92%	112%
Houston	2,707	3,859	2,468	4,151	3,029	4,043	3,983	47%	3%
Indianapolis	595	805	346	782	680	1,125	1,746	193%	117%
Jacksonville	1,852	1,486	899	1,958	1,148	1,893	1,538	-17%	3%
Kansas City, MO	233	257	50	119	558	396	630	170%	145%
Las Vegas	883	814	866	759	1,646	534	847	-4%	4%
Long Beach	1,959	1,351	1,261	1,016	1,911	2,124	2,210	13%	64%
Los Angeles	9,607	9,052	9,821	10,664	11,081	16,147	17,345	81%	92%
Louisville	211	489	658	347	753	1,081	1,366	547%	179%
Memphis	352	304	214	403	864	338	64	-82%	-79%
Mesa	1,898	2,240	1,485	1,285	3,137	1,511	2,220	17%	-1%
Miami	895	700	783	445	187	710	811	-9%	16%
Milwaukee	903	833	1,290	1,154	1,629	2,890	1,675	85%	101%
Minneapolis	3,014	3,856	4,589	4,835	7,198	8,164	8,036	167%	108%
Nashville	361	386	448	355	659	1,261	322	-11%	-17%
New Orleans	1,689	2,187	1,712	1,500	1,672	1,183	3,742	122%	71%
New York	9,643	15,024	16,468	19,953	26,243	24,428	22,619	135%	51%
Oakland	1,758	2,085	2,529	3,690	2,278	3,711	4,884	178%	134%
Oklahoma City	291	266	422	876	217	510	222	-24%	-17%
Omaha	243	269	217	555	479	170	498	105%	85%
Philadelphia	3,637	4,908	4,778	6,403	5,753	9,410	13,304	266%	171%
Phoenix	5,168	5,146	3,940	4,175	3,751	5,986	6,184	20%	20%
Portland, OR	2,453	4,775	8,942	11,477	10,987	17,365	16,846	587%	253%
Raleigh	510	508	540	526	722	610	1,355	166%	167%
Sacramento	2,971	2,252	3,305	2,455	3,710	5,658	4,090	38%	82%
San Antonio	593	788	669	447	822	664	859	45%	9%
San Diego	6,111	4,214	3,602	4,981	5,340	5,667	5,212	-15%	24%
San Francisco	3,634	8,302	7,053	8,938	10,514	12,038	13,023	258%	57%
San Jose	2,486	2,638	1,622	1,903	3,033	5,531	3,908	57%	48%
Seattle	4,179	5,943	6,963	7,330	7,336	9,953	10,593	153%	78%
Tucson	4,957	4,791	5,230	3,774	4,393	5,029	4,439	-10%	-7%
Tulsa	361	385	456	113	201	784	1,132	214%	194%
Virginia Beach	912	719	1,230	1,240	526	761	2,160	137%	200%
Washington, DC	2,292	3,035	4,336	5,667	4,871	7,066	6,306	175%	108%
Total/Average	100,990	121,514	133,119	153,258	167,136	207,015	219,192	105%(1)	80%(1)
Median	1,352	1,242	1,261	1,285	1,646	1,620	2,220	-0.8%	72%
High	9,643	15,024	16,468	19,953	26,243	24,428	22,619	587%	386%
Low	211	246	50	113	133	170	64	-89%	-79%

Legend:
= High value
= Low value

Sources: U.S. Census 1990, 2000; ACS 2005-2009 **Note:** (1) Average.

Bicycle to Work Levels by State 1990-2009

State	# of daily bicycle commuters							% Change '90-'09	% Change '00-'09
	1990	2000	2005	2006	2007	2008	2009		
Alabama	1,781	1,414	1,814	1,315	2,027	2,632	2,392	34%	69%
Alaska	1,717	1,583	2,629	2,604	3,370	2,806	2,689	57%	70%
Arizona	22,134	22,209	20,014	20,412	21,407	25,094	26,015	18%	17%
Arkansas	1,022	1,511	2,040	1,953	1,510	1,611	1,516	48%	0%
California	130,706	120,567	109,912	128,960	144,406	167,190	158,477	21%	31%
Colorado	13,140	16,905	21,344	25,686	27,014	31,893	34,500	163%	104%
Connecticut	2,855	2,875	4,398	4,557	4,689	4,100	5,072	78%	76%
Delaware	1,131	851	1,186	1,775	1,572	1,240	1,255	11%	47%
Florida	40,726	39,294	33,705	43,651	41,649	49,775	51,684	27%	32%
Georgia	4,807	5,588	6,543	6,835	8,077	9,192	10,158	111%	82%
Hawaii	6,100	4,888	4,057	4,758	3,694	6,265	8,128	33%	66%
Idaho	3,509	3,942	6,589	5,133	7,626	10,830	7,831	123%	99%
Illinois	13,922	18,406	22,126	28,073	32,313	33,346	33,449	140%	82%
Indiana	6,150	7,725	11,363	11,164	9,407	13,187	12,069	96%	56%
Iowa	4,369	5,244	7,992	7,321	6,192	7,572	5,797	33%	11%
Kansas	3,181	2,966	3,567	4,624	4,375	5,598	5,369	69%	81%
Kentucky	1,595	2,609	2,062	2,389	4,016	3,578	4,129	159%	58%
Louisiana	6,089	6,648	5,064	6,428	5,525	7,059	8,056	32%	21%
Maine	1,455	1,402	1,801	2,652	2,435	3,436	3,202	120%	128%
Maryland	4,715	4,843	4,744	7,545	5,006	9,327	10,315	119%	113%
Massachusetts	11,285	12,355	11,967	16,778	18,803	23,672	26,832	138%	117%
Michigan	9,196	10,034	12,294	15,263	15,487	20,046	19,689	114%	96%
Minnesota	8,450	10,096	13,766	16,660	17,838	24,009	18,968	124%	88%
Mississippi	1,519	1,112	1,318	1,541	3,157	2,948	995	-34%	-11%
Missouri	2,941	3,937	5,003	4,920	6,115	5,898	6,936	136%	76%
Montana	3,209	4,049	7,296	6,048	6,535	6,916	7,946	148%	96%
Nebraska	2,814	2,547	3,558	4,366	5,227	4,466	4,126	47%	62%
Nevada	4,483	4,545	4,533	6,490	6,862	5,698	4,811	7%	6%
New Hampshire	1,721	1,218	1,355	1,584	2,235	2,263	2,608	52%	114%
New Jersey	9,183	9,142	10,596	13,382	11,834	13,272	13,239	44%	45%
New Mexico	4,389	4,287	4,218	4,795	4,025	9,076	5,870	34%	37%
New York	20,159	25,036	28,987	36,279	41,879	43,186	39,185	94%	57%
North Carolina	7,136	6,840	7,947	10,172	8,383	10,388	10,766	51%	57%
North Dakota	1,030	1,011	1,397	1,703	1,943	2,853	1,353	31%	34%
Ohio	7,703	9,535	12,050	10,938	15,679	18,320	13,780	79%	45%
Oklahoma	2,721	2,910	3,641	3,157	2,942	4,449	3,964	46%	36%
Oregon	13,647	17,172	25,477	28,979	32,937	37,582	39,920	193%	132%
Pennsylvania	12,556	14,001	14,560	18,092	18,361	24,381	29,316	133%	109%
Rhode Island	1,041	1,338	1,140	1,194	1,203	1,521	1,978	90%	48%
South Carolina	4,598	3,874	4,270	5,340	3,830	4,579	5,819	27%	50%
South Dakota	1,258	950	1,581	2,760	2,014	2,164	2,353	87%	148%
Tennessee	1,818	2,330	1,916	2,697	3,226	4,592	3,004	65%	29%
Texas	18,460	21,551	22,938	23,514	25,483	29,309	26,211	42%	22%
Utah	5,010	5,267	7,103	7,567	9,806	9,238	10,584	111%	101%
Vermont	1,054	977	1,635	1,497	1,579	2,143	2,550	142%	161%
Virginia	9,068	7,930	8,366	8,243	10,797	15,839	12,571	39%	59%
Washington	13,170	16,205	19,255	21,790	21,491	28,974	28,395	116%	75%
West Virginia	530	755	506	872	1,635	1,209	1,208	128%	60%
Wisconsin	11,802	11,635	17,264	20,066	19,062	21,471	20,009	70%	72%
Wyoming	1,509	1,353	1,673	2,850	3,310	2,839	2,308	53%	71%
Total/Average	466,856	488,497	534,896	623,039	664,859	786,098	765,703	64%(1)	57%(1)
Median	4,598	4,843	5,064	4,628	6,192	7,316	7,889	69%	64%
High	130,706	120,567	109,912	128,960	144,406	167,190	158,477	193%	161%
Low	530	755	506	872	1,203	1,209	995	-34%	-11%

Sources: U.S. Census 1990, 2000; ACS 2005-2009 **Note:** (1) Average.

Legend:
= High value
= Low value

Walking to Work Levels by City 1990-2009

City	# of daily walking commuters							% Change '90-'09	% Change '00-'09
	1990	2000	2005	2006	2007	2008	2009		
Albuquerque	5,358	5,785	5,173	5,274	5,938	5,829	3,981	-26%	-31%
Arlington, TX	2,428	2,761	1,425	1,836	2,602	4,014	3,430	41%	24%
Atlanta	6,453	6,261	6,068	9,350	7,922	9,643	11,450	77%	83%
Austin	8,058	8,995	6,374	7,901	8,099	7,967	9,677	20%	8%
Baltimore	22,906	17,727	13,819	20,549	18,302	16,816	19,134	-16%	8%
Boston	39,450	36,323	31,769	39,913	40,598	45,522	47,840	21%	32%
Charlotte	4,623	4,269	4,762	5,712	5,973	6,130	8,460	83%	98%
Chicago	76,041	67,556	63,580	64,866	67,084	73,472	75,469	-1%	12%
Cleveland	8,964	7,080	6,471	7,133	5,726	7,221	7,208	-20%	2%
Colorado Springs	4,370	4,514	4,661	3,760	4,841	4,858	5,232	20%	16%
Columbus	13,494	11,743	5,528	10,017	9,406	10,830	9,879	-27%	-16%
Dallas	12,050	10,466	9,675	10,400	8,089	11,084	11,406	-5%	9%
Denver	12,345	12,112	12,967	11,448	12,434	13,131	11,300	-8%	-7%
Detroit	10,919	8,977	6,759	8,457	6,793	6,125	11,670	7%	30%
El Paso	5,917	4,075	4,531	5,020	5,303	4,712	6,339	7%	56%
Fort Worth	4,627	4,036	3,004	5,777	3,500	3,766	3,894	-16%	-4%
Fresno	3,732	3,222	3,094	4,052	3,951	3,613	3,464	-7%	8%
Honolulu	12,494	11,404	12,004	11,633	11,725	15,111	17,058	37%	50%
Houston	23,194	19,413	16,357	22,455	20,901	19,762	24,318	5%	25%
Indianapolis	8,825	7,705	6,722	6,656	6,237	8,919	7,376	-16%	-4%
Jacksonville	8,372	6,271	6,545	6,506	4,934	6,567	6,447	-23%	3%
Kansas City, MO	5,838	4,731	4,796	4,255	4,584	3,902	5,297	-9%	12%
Las Vegas	4,634	4,545	4,541	4,978	5,326	3,838	6,288	36%	38%
Long Beach	6,185	4,674	3,766	5,244	7,468	7,285	4,597	-26%	-2%
Los Angeles	63,885	53,386	52,416	58,869	64,134	61,819	59,805	-6%	12%
Louisville	4,346	4,539	3,426	4,273	6,219	6,621	5,456	26%	20%
Memphis	6,569	5,300	5,508	5,575	5,631	6,099	5,031	-23%	-5%
Mesa	3,322	3,794	4,083	4,666	4,207	4,192	4,000	20%	5%
Miami	6,144	4,646	7,203	3,870	6,269	6,163	6,378	4%	37%
Milwaukee	16,051	11,770	9,586	12,776	11,516	10,689	12,479	-22%	6%
Minneapolis	14,798	13,488	11,004	13,735	12,169	11,592	13,308	-10%	-1%
Nashville	6,485	6,509	4,815	5,758	3,620	6,732	4,282	-34%	-34%
New Orleans	9,762	9,822	7,479	3,915	7,055	8,202	8,736	-11%	-11%
New York	340,077	332,264	323,712	355,154	378,073	392,786	384,065	13%	16%
Oakland	7,787	6,355	4,898	7,970	8,379	7,735	7,037	-10%	11%
Oklahoma City	4,093	3,714	3,316	4,398	2,539	4,341	3,841	-6%	3%
Omaha	5,445	4,659	3,952	5,279	3,925	5,242	7,559	39%	62%
Philadelphia	66,446	51,564	43,259	44,102	45,003	49,590	53,533	-19%	4%
Phoenix	12,874	12,998	10,730	12,991	12,383	10,921	13,464	5%	4%
Portland, OR	12,058	14,192	11,076	14,264	12,232	15,482	16,125	34%	14%
Raleigh	4,087	4,383	2,913	3,549	5,768	4,956	5,040	23%	15%
Sacramento	5,416	4,602	6,905	5,586	6,888	7,336	5,556	3%	21%
San Antonio	12,244	10,679	7,873	13,614	12,451	10,839	12,367	1%	16%
San Diego	27,250	21,172	10,938	22,632	16,465	18,821	18,364	-33%	-13%
San Francisco	37,611	39,192	36,629	37,934	40,241	41,621	45,227	20%	15%
San Jose	6,495	6,170	6,131	8,183	8,645	8,142	8,480	31%	37%
Seattle	20,250	23,291	20,737	26,686	26,907	31,419	27,249	35%	17%
Tucson	7,237	7,438	7,256	9,942	9,434	8,098	8,567	18%	15%
Tulsa	4,995	4,195	3,440	3,504	4,510	4,429	3,475	-30%	-17%
Virginia Beach	7,373	4,369	3,429	8,257	4,621	5,874	3,325	-55%	-24%
Washington, DC	35,978	30,785	24,905	33,625	32,163	36,636	32,328	-10%	5%
Total/Average	1,064,763	969,921	882,010	1,014,299	1,029,183	1,086,494	1,096,291	3%(1)	13%(1)
Median	8,058	7,080	6,545	7,970	7,468	7,967	8,480	-1%	11%
High	340,077	332,264	323,712	355,154	378,073	392,786	384,065	83%	98%
Low	2,428	2,761	1,425	1,836	2,539	3,613	3,325	-55%	-34%

Sources: U.S. Census 1990, 2000; ACS 2005-2009 **Note:** (1) Average.

Legend:
= High value
= Low value

Walking to Work Levels by State 1990-2009

State	# of daily walking commuters							% Change '90-'09	% Change '00-'09
	1990	2000	2005	2006	2007	2008	2009		
Alabama	32,873	25,360	23,230	22,003	25,887	25,826	25,810	-21%	2%
Alaska	26,927	21,298	19,368	28,841	27,750	24,288	26,751	-1%	26%
Arizona	54,648	58,015	54,084	63,952	62,419	55,038	65,624	20%	13%
Arkansas	27,058	21,915	19,301	21,851	19,651	25,614	22,866	-15%	4%
California	469,867	414,581	384,989	440,072	462,555	470,877	447,411	-5%	8%
Colorado	69,041	65,668	66,273	73,495	77,613	69,642	73,437	6%	12%
Connecticut	61,484	44,348	33,775	52,221	53,229	49,997	50,342	-18%	14%
Delaware	12,862	9,637	7,159	10,865	11,001	9,795	9,713	-24%	1%
Florida	145,269	118,386	121,370	137,621	134,794	125,397	118,132	-19%	0%
Georgia	72,640	65,776	54,467	74,829	71,522	66,106	70,992	-2%	8%
Hawaii	31,935	27,134	19,369	30,287	27,727	27,763	29,016	-9%	7%
Idaho	20,091	20,747	21,841	23,073	21,645	22,640	17,568	-13%	-15%
Illinois	225,942	180,119	151,285	171,224	183,864	185,226	189,926	-16%	5%
Indiana	84,324	69,184	54,512	65,675	61,439	69,561	62,950	-25%	-9%
Iowa	76,572	58,088	46,469	56,101	58,839	64,035	61,390	-20%	6%
Kansas	45,346	33,271	33,761	36,467	37,420	37,168	38,761	-15%	17%
Kentucky	54,938	42,494	30,024	37,637	40,320	45,380	43,922	-20%	3%
Louisiana	48,216	40,184	32,101	32,165	37,577	38,373	39,371	-18%	-2%
Maine	30,813	24,700	23,204	27,536	26,257	25,976	26,414	-14%	7%
Maryland	83,417	64,852	56,401	73,327	71,127	68,399	74,434	-11%	15%
Massachusetts	161,820	134,566	114,505	133,638	136,920	151,996	151,189	-7%	12%
Michigan	125,501	101,506	79,324	99,422	101,010	97,835	98,309	-22%	-3%
Minnesota	105,328	84,148	70,164	83,377	82,058	81,952	77,059	-27%	-8%
Mississippi	27,142	21,868	15,274	21,646	20,750	21,662	19,684	-27%	-10%
Missouri	66,553	55,631	48,382	57,187	55,818	57,211	54,127	-19%	-3%
Montana	27,022	23,336	21,160	24,302	21,982	26,069	24,724	-9%	6%
Nebraska	36,914	28,003	23,631	31,898	27,182	29,332	30,642	-17%	9%
Nevada	24,866	24,875	26,044	25,553	28,654	29,787	25,187	1%	1%
New Hampshire	23,137	18,545	18,381	23,634	22,824	23,365	19,158	-17%	3%
New Jersey	156,523	121,305	122,068	141,051	133,200	142,567	137,504	-12%	13%
New Mexico	21,923	21,435	21,064	19,117	19,460	23,238	20,989	-4%	-2%
New York	575,089	511,721	469,473	547,956	558,152	583,118	574,322	0%	12%
North Carolina	96,614	74,147	53,946	73,056	78,373	75,088	82,681	-14%	12%
North Dakota	24,111	16,094	13,492	13,505	14,814	12,925	12,419	-48%	-23%
Ohio	156,648	125,882	97,639	134,493	121,594	118,969	116,395	-26%	-8%
Oklahoma	39,782	32,796	27,643	32,813	31,494	33,566	30,444	-23%	-7%
Oregon	53,953	57,217	54,015	68,134	62,702	70,638	65,863	22%	15%
Pennsylvania	304,589	229,725	201,199	234,674	228,848	235,564	227,700	-25%	-1%
Rhode Island	20,727	18,717	12,701	15,426	16,929	15,576	15,797	-24%	-16%
South Carolina	50,538	42,567	27,136	35,316	35,438	37,212	36,672	-27%	-14%
South Dakota	22,578	16,786	15,277	17,092	18,828	19,488	17,914	-21%	7%
Tennessee	50,773	39,689	36,376	39,065	39,075	39,283	38,131	-25%	-4%
Texas	202,494	173,670	148,535	195,559	189,007	196,347	192,756	-5%	11%
Utah	25,080	28,523	23,991	32,864	32,736	38,437	36,985	47%	30%
Vermont	19,001	17,554	16,563	19,650	20,312	21,088	17,214	-9%	-2%
Virginia	97,766	80,487	63,042	87,750	81,547	89,107	85,818	-12%	7%
Washington	91,475	89,739	83,595	99,717	106,144	113,874	103,950	14%	16%
West Virginia	29,511	21,059	18,135	23,118	18,828	23,802	20,652	-30%	-2%
Wisconsin	130,136	100,301	79,439	99,410	93,824	97,390	94,869	-27%	-5%
Wyoming	11,051	10,548	11,319	8,244	10,908	10,771	9,347	-15%	-11%
Total/Average	4,488,886	3,758,982	3,291,401	3,951,534	3,954,210	4,060,994	3,965,659	-12% (1)	5% (1)
Median	53,953	42,567	33,775	39,065	40,320	42,332	41,647	-16%	4%
High	**575,089**	**511,721**	**469,473**	**547,956**	**558,152**	**583,118**	**574,332**	**47%**	**30%**
Low	**11,051**	**9,637**	**7,159**	**8,244**	**10,908**	**9,795**	**9,347**	**-48%**	**-23%**

Sources: U.S. Census 1990, 2000; ACS 2005-2009 **Note:** (1) Average.

Legend:
= High value
= Low value

Appendix 5: Additional Resources

Advocacy Organizations:

State and Local Advocacy Organizations:

• See www.PeoplePoweredMovement.org to find your state or local bicycle and pedestrian advocacy organization

National Advocacy Organizations:

- **Adventure Cycling Association**: http://www.adventurecycling.org
- **Alliance for Biking & Walking:** http://www.PeoplePoweredMovement.org
- **America Bikes:** http://www.americabikes.org
- **American Public Health Association:** http://bit.ly/d5iw6O
- **American Trails:** http://www.americantrails.org/
- **America Walks:** http://www.americawalks.org
- **Association of Pedestrian and Bicycle Professionals:** http://www.apbp.org
- **Bikes Belong Coalition**: http://www.bikesbelong.org
- **International Mountain Bicycling Association:** http://www.imba.com
- **League of American Bicyclists**: http://www.bikeleague.org
- **National Center for Bicycling and Walking:** http://www.bikewalk.org
- **National Complete Streets Coalition:** http://www.completestreets.org
- **Rails-to-Trails Conservancy**: http://www.railstotrails.org
- **Safe Routes to School National Partnership:** http://www.saferoutespartnership.org

Economic Impact:

- **The Hidden Health Costs of Transportation:** http://bit.ly/cMo7HI
- **Economic Benefits of Bicycle Infrastructure:** http://bit.ly/rchKVd
- **Economic Impact of Road Riding Events:** http://bit.ly/qUTTrG
- **How Bicycling Investments Affect Real Estate:** http://bit.ly/p7I9SI
- **Economic Value Walkability:** http://www.vtpi.org/walkability.pdf
- **Economic Impact Federal Investment in Bicycling: 10 Case Studies:** http://bit.ly/funydz
- **Economic Impact of U.S. Bike Route:** http://bit.ly/qAsHAq
- **Health Economic Assessment Tool (World Health Organization):** http://www.heatwalkingcycling.org/

Economic Impact Studies

- **Baltimore:** www.bikeleague.org/resources/reports/pdfs/baltimore_Dec20.pdf
- **Colorado:** http://www.atfiles.org/files/pdf/CObikeEcon.pdf
- **Florida, California, and Iowa (trails):** http://bit.ly/qaepVb
- **Maine:** http://www.maine.gov/mdot/opt/pdf/biketourismexecsumm.pdf
- **Maryland/Pennsylvania:** http://bit.ly/pSIQKg
- **Minnesota:** http://bit.ly/c1YuLK
- **New York (trails):** http://nysparks.com/recreation/trails/statewide-plans.aspx.; Statewide Trails Plan. Appendix C – Every Mile Counts – An Analysis of the 2008 Trail User Surveys.
- **North Carolina (Outer Banks):** http://1.usa.gov/oZLo5n
- **Portland (cost:benefit):** http://bit.ly/nC5nY9
- **Portland (bike industry):** http://bit.ly/kMQih4
- **San Francisco (bike lanes):** http://www.emilydrennen.org/TrafficCalming_full.pdf
- **Virginia (trail):** http://atfiles.org/files/pdf/VACstudy04.pdf
- **Wisconsin:** http://bit.ly/aMnH09

Education:

- **Blueprint for a Bicycle Friendly America:** http://bit.ly/nfrDDg
- **State Bike Summit Guide:** http://bit.ly/qYd1jg

Share the Road:

- **Colorado (3-2-1 Courtesy Code):** http://bicyclecolo.org/page.cfm?PageID=1030
- **Maine (Share the Road):** http://www.sharetheroadmn.org/resources.html
- **Minnesota (Share the Road):** http://www.sharetheroadmn.org
- **New York City (Give Respect/Get Respect):** http://bit.ly/6tp1C
- **San Francisco (Coexist):** http://www.sfbike.org/?coexist
- **South Carolina (Safe Streets Save Lives):** http://www.safestreetssavelives.org
- **South Carolina (Share the Road):** http://www.pccsc.net/sharetheroad.php

Model Bicycle Education Programs:

- **Arizona Bike Safety Classes:** http://www.dot.pima.gov/tpcbac/SafetyClasses.htm
- **Arizona Education Guides:** http://www.azbikeped.org/education.html
- **Delaware:** http://bit.ly/mBFKZ
- **Connecticut:** http://www.wecyclect.org/education/
- **Florida:** http://www.floridabicycle.org/programs/education.html
- **Hawaii:** http://www.hbl.org/content/bikeed
- **Illinois:** http://www.bikelib.org/
- **Indiana:** http://www.bicycleindiana.org/education.html
- **Kansas:** http://ksdot.org/burRail/bike/default.asp
- **Maine:** http://www.bikemaine.org/what-we-do/education
- **Michigan:** http://www.lmb.org/index.php/Education/
- **Minnesota:** http://www.bikemn.org/
- **New York:** http://www.bikenewyork.org/learn/classes-workshops/overview/
- **Oklahoma:** http://okbike.org/index.php?option=com_content&task=section&id=6&Itemid=35
- **Oregon:** http://www.bta4bikes.org/at_work/programs.php
- **Texas:** http://www.biketexas.org/en/education
- **Vermont:** http://www.vtbikeped.org/what/safety.htm
- **West Virginia:** http://www.wvcf.org/home/

Encouragement:

- **Blueprint for a Bicycle Friendly America:** http://bit.ly/nfrDDg

Bike to Work Day Events:

- **Baltimore:** http://www.baltometro.org/commuter-options/bike-to-work-day
- **Cleveland:** http://www.clevelandbicycleweek.org/events/bike-work-day
- **Denver:** http://www.drcog.org/btwd2009/
- **Louisville:** http://www.louisvilleky.gov/BikeLouisville/biketoworkday2011.htm
- **San Francisco:** http://www.sfbike.org/?btwd and http://www.youcanbikethere.com/
- **San Jose:** http://bikesiliconvalley.org/btwd
- **Washington, DC:** http://www.waba.org/events/btwd/

Open Streets/Ciclovias/Sunday Parkways:

- **See the current info on over 60 open streets initiatives at:** http://www.OpenStreetsProject.org
- **Baltimore:** http://www.baltimorespokes.org/article.php?story=20070821100331287
- **Chicago:** http://www.activetrans.org/openstreets
- **Los Angeles:** http://www.ciclavia.org/

- **Miami:** http://bikemiamiblog.wordpress.com/
- **New York:** http://www.nyc.gov/html/dot/summerstreets/html/home/home.shtml
- **Oakland:** http://oaklavia.org/
- **Portland:** http://www.portlandonline.com/Transportation/index.cfm?c=46103
- **San Francisco:** http://sundaystreetssf.com/
- **Seattle:** http://www.seattle.gov/transportation/summerstreets.htm

Promotional Rides:

- **Chicago's Bike the Drive:** http://www.bikethedrive.org
- **Iowa's Register's Annual Great Bicycle Ride Across Iowa:** http://ragbrai.com
- **Louisville's Mayor's Healthy Hometown Hike and Bike:** http://1.usa.gov/o2cPn5

Public Bike Sharing:

- **Chicago:** http://chicago.bcycle.com/
- **Denver:** http://www.denverbikesharing.org/
- **Minneapolis:** https://www.niceridemn.org/
- **Nashville:** http://www.nashvillebikeshare.org/
- **Washington, DC:** http://www.capitalbikeshare.com/

Engineering

- **Blueprint for a Bicycle Friendly America:** http://bit.ly/nfrDDg

Bicycle Parking:

- **APBP's Bicycle Parking Guidelines**: http://www.apbp.org/?page=Publications
- **Minneapolis:** http://www.ci.minneapolis.mn.us/bicycles/bikeparking.asp
- **Stolen Bicycle Registry:** http://www.stolenbicycleregistry.com

Bicycle and Pedestrian Facility Design:

- **Bicycle Facility Design**: http://www.bicyclinginfo.org/engineering/
- **Outdoor Developed Areas (recreational trails):** http://www.access-board.gov/outdoor/index.htm
- **Pedestrian Facility Design**: http://www.walkinginfo.org/engineering/
- **Public Rights of Way:** http://www.access-board.gov/prowac/index.htm
- **Shared Use Paths:** http://www.access-board.gov/sup.htm
- **Urban Bikeway Design Guide:** http://nacto.org/cities-for-cycling/design-guide/

Funding:

- **America Bikes Funding Fact Sheet:** http://bit.ly/pkwbNQ
- **Congestion Mitigation and Air Quality Improvement Program:** http://bit.ly/r15wxM
- **Highway Safety Improvement Program:** http://bit.ly/r8vwB8
- **Highway Safety Improvement Program Case Studies:** http://bit.ly/pnKSLG
- **Federal Funding for Bicycling:** http://1.usa.gov/qXGI0K
- **Rescissions:** http://www.advocacyadvance.org/site_images/content/Rescissions_FAQs.pdf
- **Transportation Enhancements:** http://www.enhancements.org

Infrastructure:

- **Bicycles on Bridges**: http://www.advocacyadvance.org/docs/Bridge_Access_Report.pdf
- **Economic Benefits of Bicycle Infrastructure:** http://bit.ly/rchKVd

Sharrows:

- **San Francisco:** http://www.sfmta.com/cms/bproj/22747.html
- **Seattle:** http://www.cityofseattle.net/transportation/sharrows.htm

Environment

Climate Change/Air Quality

- **Climate Change and Bicycling:** http://bit.ly/bbe92z
- **Congestion Mitigation and Air Quality Improvement Program:** http://bit.ly/r15wxM

Healthy and Active Living:

- **American Public Health Association:** http://bit.ly/d5iw6O
- **Active Living Research:** http://www.activelivingresearch.org/
- **Centers for Disease Control and Prevention**: http://www.cdc.gov/HealthyYouth/index.htm
- **Health Economic Assessment Tool:** http://www.heatwalkingcycling.org/
- **Healthy Places (CDC):** www.cdc.gov/healthyplaces
    - Fact Sheets: http://www.cdc.gov/healthyplaces/factsheets.htm
    - Healthy Community Design: http://www.cdc.gov/healthyplaces/healthy_comm_design.htm
    - Health Impact Assessment: http://www.cdc.gov/healthyplaces/hia.htm
    - Images: http://www.cdc.gov/healthyplaces/images.htm
    - Increasing Physical Activity: http://www.cdc.gov/healthyplaces/healthtopics/physactivity.htm
    - Reducing Injury: http://www.cdc.gov/healthyplaces/healthtopics/injury.htm
- **Kaiser Permanente's Thrive Campaign:** http://thrivewithkp.org/
- **National Environmental Public Health Tracking:** http://bit.ly/mAF7cT
- **Robert Woods Johnson Foundation Active Living by Design**: http://www.activelivingbydesign.org

Health Impact Assessments:

- **Oregon (Crook County):** http://1.usa.gov/p0hUcx
- **Sacramento:** http://www.ph.ucla.edu/hs/health-impact/docs/WalktoschoolSummary.pdf
- **Washington (Clark County):** http://bit.ly/r54yTu

International Organizations

- **Denmark Cycling Embassy:** http://www.cycling-embassy.dk/
- **European Cyclists Federation:** http://www.ecf.com/
- **Fietsberaad:** http://www.fietsberaad.nl/

Maps:

- **Arizona Bicycle Maps:** http://www.dot.pima.gov/tpcbac/Publications.html#map
  and http://www.azbikeped.org/maps.htm
- **Colorado:** http://bicyclecolo.org/page.cfm?PageID=626
- **Delaware:** http://bit.ly/2yvA13
- **Denver:** http://www.bikedenver.org/maps/
- **Illinois:** http://www.dot.state.il.us/bikemap/STATE.HTML
- **Louisville:** http://www.louisvilleky.gov/BikeLouisville/IWantTo/existingbikelanes.htm
- **Maine:** http://www.exploremaine.org/bike/search-bike.shtml
- **Michigan:** http://bit.ly/caNrl
- **Milwaukee:** http://www.ci.mil.wi.us/maps4460.htm

- **Minneapolis:** http://www.ci.minneapolis.mn.us/bicycles/where-to-ride.asp
- **Minnesota:** http://www.dot.state.mn.us/bike/
- **New Hampshire:** http://www.nh.gov/dot/programs/bikeped/maps/index.htm
- **New Jersey:** http://www.njbikemap.com/
- **New York:** http://www.nycbikemaps.com/
- **North Carolina:** http://www.ncdot.org/it/gis/DataDistribution/BikeMaps/
- **Ohio:** http://www.noaca.org/bikemaps.html
- **Oklahoma:** http://www.oklahomabicyclesociety.com/Maps/maphome.htm
- **Oregon:** http://www.oregon.gov/ODOT/HWY/BIKEPED/maps.shtml
- **Philadelphia:** http://www.bicyclecoalition.org/resources/maps
- **Portland:** http://bit.ly/lEzWp
- **San Francisco:** http://www.sfbike.org/?maps
- **Seattle:** http://www.cityofseattle.net/transportation/bikemaps.htm
- **Washington, DC:** http://www.waba.org/resources/maps.php
- **Wisconsin:** http://www.dot.wisconsin.gov/travel/bike-foot/bikemaps.htm

Master Plans:

Bicycle & Pedestrian Master Plans:

- **Arizona:** http://www.azbikeped.org/statewide-bicycle-pedestrian.html
- **Arlington, TX:** http://www.arlingtontx.gov/planning/HikeandBike.html
- **Atlanta:** http://bit.ly/pmYIYp
- **Las Vegas:** http://www.rtcsouthernnevada.com/mpo/plansstudies/nmamp/index.cfm
- **Nashville:** http://mpw.nashville.gov/IMS/stratplan/PlanDownload.aspx
- **Sacramento:** http://www.sacog.org/bikeinfo/download_bike_ped_trails_mp.cfm

Bicycle Master Plans:

- **Austin:** http://www.ci.austin.tx.us/publicworks/bicycle-plan.htm
- **Baltimore:** http://1.usa.gov/rkgyvT
- **Chicago:** http://bike2015plan.org/
- **Columbus:** http://www.altaprojects.net/columbus/
- **Dallas:** http://www.tooledesign.com/dallasbikeplan/
- **Delaware:** http://bit.ly/1qfa1T
- **Denver:** http://bit.ly/kH56Mf
- **Fresno:** http://bit.ly/11E7HM
- **Hawaii:** http://hawaii.gov/dot/highways/Bike/Bike%20Plan/index.htm
- **Honolulu:** http://www1.honolulu.gov/dts/bikeway/
- **Los Angeles:** http://www.labikeplan.org/
- **Long Beach:** http://bit.ly/vFOTi
- **Louisville:** http://www.louisvilleky.gov/BikeLouisville/
- **Minneapolis:** http://www.ci.minneapolis.mn.us/bicycles/bicycle-plans.asp
- **Nevada:** http://www.bicyclenevada.com/bikeplan03.htm
- **New York City:** http://www.nyc.gov/html/dcp/html/bike/mp.shtml
- **Oakland:** http://bit.ly/njCGb2
- **Portland, OR:** http://bit.ly/17AeXX
- **Raleigh:** http://1.usa.gov/nAnBFG
- **Sacramento County:** http://bit.ly/nq5yst
- **San Diego:** http://bit.ly/1271Kl
- **San Francisco:** http://www.sfmta.com/cms/bproj/bikeplan.htm
- **Seattle:** http://www.cityofseattle.net/transportation/bikemaster.htm

Pedestrian Master Plans:

- **Austin:** http://www.ci.austin.tx.us/publicworks/downloads/sidewalk_mp_resolution.pdf
- **Kansas City:** http://bit.ly/p8E8hn
- **Louisville:** http://www.louisvilleky.gov/HealthyHometown/StepUpLouisville/pedmasterplan/
- **Minneapolis:** http://bit.ly/TFTqB
- **Oakland:** www.oaklandnet.com/government/pedestrian/PedMasterPlan.pdf
- **Portland, OR:** http://www.portlandonline.com/transportation/index.cfm?c=34778&a=292295
- **San Diego:** http://bit.ly/WsW5r
- **San Francisco:** http://www.sfmta.com/cms/wproj/28717.html
- **Seattle:** http://www.seattle.gov/transportation/pedestrian_masterplan
- **Washington, DC:** http://1.usa.gov/nHvOL9

Policies:

Advisory Committees:

- **Arlington Bicycle Advisory Committee:** http://www.nctcog.org/trans/committees/bpac/index.asp
- **California Bicycle Advisory Committee:** http://www.dot.ca.gov/hq/tpp/offices/bike/cbac.html
- **City of Columbus Bikeway Advisory Committee:** http://bit.ly/nRJhhi
- **Denver Bicycling Advisory Committee:** http://bit.ly/QUqTZ
- **Fresno Bicycle/Pedestrian Advisory Committee:** http://bit.ly/1yWDmp
- **Fort Worth Bicycle and Pedestrian Advisory Committee:** http://bit.ly/4oErlO
- **Houston Pedestrian and Bicycle Advisory Committee:** http://bit.ly/ocFpQX
- **Los Angeles Bicycle Advisory Committee:** http://ladot.lacity.org/tf_Bicycle_advisory.htm
- **Los Angeles Pedestrian Advisory Committee:** http://bit.ly/chcHn2
- **Maryland Bicycle and Pedestrian Advisory Committee:** http://bit.ly/jcQ1Q
- **Miami-Dade Bicycle Pedestrian Advisory Committee:** http://1.usa.gov/nyOJO3
- **Minneapolis Bicycle Advisory Committee:** http://bit.ly/1a4qt4
- **Nashville Bicycle Pedestrian Advisory Committee:** http://bit.ly/4wYc6A
- **Nevada:** http://www.bicyclenevada.com/board.html
- **Oakland Bicycle Advisory Committee:** http://bit.ly/oXsjIh
- **Omaha Bicycle Advisory Committee:** http://bit.ly/o9llI
- **San Antonio Bicycle Mobility Advisory Committee:** http://bit.ly/7tSSsv
- **San Francisco Bicycle Advisory Committee:** http://www.sfgov.org/site/bac_index.asp?id=11525
- **San Jose Bicycle Pedestrian Advisory Committee:** http://1.usa.gov/nsRXqC
- **Tucson Bicycle Advisory Committee:** http://www.dot.pima.gov/tpcbac/

Complete Streets:

- **Advice on complete streets campaigns:** http://www.PeoplePoweredMovement.org/contact
- **The latest complete streets news:** http://www.completestreets.org

Complete Streets Policies:

- **Guide to Complete Streets Campaigns:** http://www.peoplepoweredmovement.org/publications
- **Examples of Complete Streets Policies and Guides:** http://bit.ly/5Iy15q
- **Federal policy:** http://www.dot.gov/affairs/2010/bicycle-ped.html
- **California:** http://www.completestreets.org/webdocs/policy/cs-ca-legislation.pdf
- **Connecticut:** http://www.completestreets.org/webdocs/policy/cs-ct-legislation.pdf
- **Delaware:** http://governor.delaware.gov/orders/exec_order_06.shtml#TopOfPage
- **Hawaii:** http://www.completestreets.org/webdocs/policy/cs-hi-legislation.pdf
- **Illinois:** http://www.completestreets.org/webdocs/policy/cs-il-legislation.pdf
- **Louisiana:** http://www.completestreets.org/webdocs/policy/cs-la-resolution.pdf

- **Louisville:** http://www.louisvilleky.gov/BikeLouisville/Complete+Streets/
- **Massachusetts:** http://bit.ly/pVDsBQ
- **Minnesota:** http://www.completestreets.org/webdocs/policy/cs-mn-legislation.pdf
- **New Jersey:** http://www.completestreets.org/webdocs/policy/cs-nj-dotpolicy.pdf
- **North Carolina:** http://www.completestreets.org/webdocs/policy/cs-nc-dotpolicy.pdf
- **Oregon:** http://www.completestreets.org/webdocs/policy/cs-or-legislation.pdf
- **Rhode Island:** http://www.rilin.state.ri.us/statutes/title31/31-18/31-18-21.HTM
- **Wisconsin:** http://www.completestreets.org/webdocs/policy/cs-wi-legislation.pdf

Police on Bicycles:
- **International Police Mountain Biking Association:** http://www.ipmba.org

Safe Passing Laws:
- **3FeetPlease.com:** http://www.3feetplease.com/
- **Arizona:** http://azbikelaw.org/articles/ThreeFoot.html
- **Austin:** http://bit.ly/prO7XV
- **Delaware:** http://delcode.delaware.gov/title21/c041/sc03/index.shtml
- **Louisiana:** http://www.louisiana3feet.com/
- **Georgia:** http://www.legis.ga.gov/legislation/en-US/Display.aspx?Legislation=32251
- **Maine:** http://www.mainelegislature.org/legis/statutes/29-a/title29-asec2070.html
- **New Orleans:** http://bit.ly/eVzY4
- **Oklahoma City:** http://bit.ly/46paAG
- **Tennessee:** http://www.tennessee3feet.org/

Mandatory Helmet Laws:
- **Bicycle Helmet Safety Institute:** http://www.helmets.org/mandator.htm
- **Arguments/Case Study Against Mandatory Bicycle Helmet Laws:** http://www.cycle-helmets.com/
- **LAB Helmet Law Position:** http://www.bikeleague.org/about/positions/helmetuse.php
- **Arguments Against Mandatory Helmet Laws:** http://www.kenkifer.com/bikepages/advocacy/mhls.htm

Staffing:
- **Why Communities & States Need Bicycle and Pedestrian Staff:** http://bit.ly/o5Kjel

Retailers/Industry:
- **Bikes Belong Coalition:** http://www.bikesbelong.org
- **National Bicycle Dealers Association:** http://www.nbda.com

Safety:
- **Distracted Driving:** http://www.advocacyadvance.org/docs/distracted_driving_league_report.pdf
- **Highway Safety Improvement Program:** http://bit.ly/r8vwB8
- **Highway Safety Improvement Program Case Studies:** http://bit.ly/pnKSLG
- **Traffic Safety Fact Sheets:** http://bit.ly/wrKo0
- **State Traffic Safety Information:** http://bit.ly/d3EzmD

Safe Routes to School:

- **Safe Routes to School National Partnership:** www.saferoutespartnership.org
- **The National Center for Safe Routes to School:** www.saferoutesinfo.org
- **Progress Reports:** http://www.saferoutesinfo.org/resources/tracking-reports.cfm
- **State of the States:** http://bit.ly/pRGIap
- **EPA School Siting Guidelines (Draft):** http://www.epa.gov/schools/siting/

Sample Safe Routes to School Programs:

- **Boston:** http://www.walkboston.org/work/safe_routes.htm
- **California:** http://saferoutescalifornia.wordpress.com/
- **Colorado:** http://www.coloradodot.info/programs/bikeped/safe-routes
- **Connecticut:** http://www.ctsaferoutes.ct.gov/
- **Delaware:** http://deldot.gov/information/community_programs_and_services/srts
- **Denver:** http://www.denvergov.org/DenverSafeRoutestoSchool/tabid/427939/Default.aspx
- **Florida:** http://www.dot.state.fl.us/Safety/SRTS_files/SRTS.shtm
- **Illinois:** http://www.dot.il.gov/saferoutes/saferouteshome.aspx
- **Indiana:** http://www.in.gov/indot/2355.htm
- **Iowa:** http://www.iowadot.gov/saferoutes/
- **Kansas:** http://www.ksdot.org/burTrafficEng/sztoolbox/default.asp
- **Louisiana:** http://www.dotd.louisiana.gov/planning/highway_safety/safe_routes/
- **Maine:** http://www.bikemaine.org/what-we-do/maine-safe-routes-to-school-program
- **Massachusetts:** http://www.commute.com/schools
- **Michigan:** http://www.saferoutesmichigan.org/
- **Minnesota:** http://www.dot.state.mn.us/saferoutes/
- **Mississippi:** http://bit.ly/1iQixg
- **Missouri:** http://www.modot.mo.gov/safety/saferoutestoschool.htm
- **Montana:** http://www.mdt.mt.gov/pubinvolve/saferoutes/
- **Nebraska:** http://www.saferoutesne.com/
- **New Jersey:** http://www.state.nj.us/transportation/community/srts/
- **New Mexico:** http://www.nmshtd.state.nm.us/main.asp?secid=15411
- **New York:** http://bit.ly/XVFMv
- **North Carolina:** http://www.ncdot.org/doh/preconstruct/traffic/congestion/CM/msta/docs/SRTS.pdf
- **Oklahoma:** http://www.okladot.state.ok.us/srts/index.php
- **Portland:** http://www.portlandonline.com/TRANSPORTATION/index.cfm?c=40511
- **South Carolina:** http://www.scdot.org/community/saferoutes.shtml
- **Texas:** http://www.saferoutestx.org/
- **Wisconsin:** http://www.dot.wisconsin.gov/localgov/aid/saferoutes.htm

Statistics/Studies:

General Information:

- **Advocacy Advance:** http://www.advocacyadvance.org
- **Alliance Benchmarking Project:** http://www.peoplepoweredmovement.org/benchmarking
- **Bikes Belong:** http://www.bikesbelong.org/statistics
- **Federal Highway Administration:** http://www.fhwa.dot.gov/environment/bikeped
- **Fietsberaad (Netherlands)**: http://www.fietsberaad.nl/index.cfm?lang=en&section=Kennisbank
- **League of American Bicyclists:** http://www.bikeleague.org/resources/reports/
- **National Highway Traffic Safety Administration Traffic Safety Fact Sheets:** http://bit.ly/wrKo0
- **Pedestrian and Bicycle Information Center:** http://www.pedbikeinfo.org

- **Rails-to-Trails Conservancy:** http://www.railstotrails.org/ourWork/advocacy/activeTransportation
- **Victoria Transport Policy Institute:** http://www.vtpi.org/
- **National Environmental Public Health Tracking:** http://bit.ly/mAF7cT

Mode Share (Bicycle and Pedestrian Counts):

- **Commuter Trends:** http://www.advocacyadvance.org/docs/acs_commuting_trends.pdf
- **National Bicycle and Pedestrian Documentation Project:** http://bikepeddocumentation.org

Trainings:

- **Action 2020 Workshops:** http://www.advocacyadvance.org/trainings
- **Membership Development Training:** http://bit.ly/2Rrx7Q
- **Safe Routes to School:** http://www.saferoutestoschools.org/Programs/Workshops.htm and http://www.saferoutesinfo.org/events-and-training/national-course
- **Winning Campaigns Trainings:** http://www.peoplepoweredmovement.org/wctraining

Appendix 6: Overview of Other Benchmarking Efforts

The Alliance for Biking & Walking's Benchmarking Project is the only focused effort to set benchmarks for bicycling and walking in the United States using data from all 50 states and the 51 largest cities. Other benchmarking efforts from abroad and within the United States have provided examples and inspiration for this project.

Benchmarking Efforts Abroad

Cycling and walking benchmarking efforts have been in place longer in many other countries than in the United States, England, Scotland, and the Netherlands all have completed benchmarking projects. More than 100 cities and 18 regions in 21 European countries have participated in **BYPAD (Bicycle Policy Audit)**, developed by an international consortium of bicycle experts as part of a European Union-funded project. Velo Mondial completed a national bicycling benchmark program with five participating countries (Czech Republic, England, Finland, Scotland, and the Netherlands) that compared bicycling policies at the national level. Another multi-nation benchmarking project is the Urban Transport Benchmarking Initiative that uses benchmarking to compare European Union cities around six transport themes (Behavioral and Social Issues in Public Transport, City Logistics, Cycling, Demand Management, Public Transport Organization and Policy, and Urban Transport for Disabled People).

Benchmarking Bicycling in the UK

One benchmarking project by the **Cyclist's Touring Club (CTC)** investigated up to 10 cities per year between 2001 and 2003. The CTC investigated bicycling policy and practice in each city including how bicycling is promoted and integrated into wider transportation plans. Participating jurisdictions completed a self-auditing questionnaire, received site visits from project staff to review the self-audit and create long-range action plans, and attended group workshops to collaborate with other jurisdictions. The CTC formulated and disseminated a comprehensive list of "Best Practices" to help each area make better plans for bicycling. These "Best Practice" resources and photographs are located in a searchable database on CTC's website.

Dutch Benchmarking Sophistication

The Dutch have sophisticated benchmarking techniques which utilize advanced technology. The **Cycle Balance**, a project of the **Dutch Cyclists Union (Fietsersbond)**, began in 1999 and aims to "stimulate local authorities to adopt a (still) better cycling policy.... The secondary objective of the project is to enhance the position and strength of the local Cyclists Union branches."

The Cycle Balance assesses 10 dimensions of local conditions for bicyclists including: directness, comfort (obstruction), comfort (road surface), attractiveness, competitiveness compared to the car, bicycle use, road safety of bicyclists, urban density, bicyclists' satisfaction, and bicycling policy on paper. To measure these 10 dimensions they use questionnaires for the municipalities, a questionnaire on bicyclists' satisfaction, data from national databases, and the Quick Scan Indicator for Cycling Infrastructure.

The Quick Scan Indicator for Cycling Infrastructure selects 12 to 16 routes at random to sample. The routes go from randomly selected houses to destinations and vice versa. Meanwhile, the project's specially designed bicycles register data such as time, distance, speed, sound, and vibrations onto a laptop computer. From these results they can determine frequency of stops, waiting time, type of road surface, maneuvers and obstacles, and use the collected data to measure the competitiveness of a bicycle. No other study surveyed uses this level of sophistication to measure environmental conditions for bicycling with a standardized methodology. In the end, Cycle Balance presents a report to the municipality with an assessment of bicycling conditions in all 10 dimensions. The Alliance looks forward to emulating their thoroughness and sophisticated techniques as the Benchmarking Project expands in scope.

Tracking Progress in Copenhagen

Copenhagen's Bicycle Account is an effort by the **City of Copenhagen** to track and assess its bicycling development. Since 1995 the city has published a report every two years that looks at the city's bicycling conditions and new initiatives as well as the way in which the Copenhageners themselves perceive bicycling facilities and safety. The most recent report from 2010 (the ninth of its kind) is based on data from telephone interviews with 1,025 randomly selected Copenhagen residents as well as data from the DTU Transport Survey of Transport Behaviour. The report allows the city to track its own progress toward increasing bicycling, bicyclists' safety, and bicyclists' satisfaction.

Tracking Bicycling in Quebec

Every five years since 1995, Velo Quebec has produced a detailed report on bicycling trends in Quebec, Canada. The report relies upon data from a survey of Quebecers, an analysis of origin-destination surveys, and a compilation of traffic counts in a dozen regions of Quebec. The most recent report in 2010 showed a significant increase in the number of adults who bicycle in Quebec and found that 1 out of 3 Quebecers bicycle enough to derive physical fitness benefits (Velo Quebec 2010).

Benchmarking Toronto against Other World Cities

In 2008, the **Toronto Coalition for Active Transportation (TCAT)** released a benchmarking report that compared Toronto's bicycling progress to other world cities. The report highlighted bicycling mode share, funding, infrastructure, and gender of bicyclists. By comparing Toronto to other world cities leading in bicycling, TCAT made the case for increased investment in bicycling. Their report is a model for other cities on how to glean information for this Benchmarking Report and other sources to highlight the strengths, weaknesses, and opportunities in regard to bicycling and walking.

Measuring Walking

Since the launch of this Benchmarking Project, **Walk 21** has launched an international effort to measure walking. **Measuring Walking** is an effort to standardize monitoring methods of walking and public space. The project is ongoing and is seeking agreement from experts from different countries as to best practice methods and guidelines.

At the same time, a number of cities are now using the Making Walking Count survey tool to better understand the nature of walking in their neighborhoods and the issues and ideas walkers have.

51 Largest Cities with "Bicycle Friendly" Status

Platinum:	Portland, OR
Gold:	Minneapolis San Francisco Seattle Tucson
Silver:	Austin Boston Chicago Colorado Springs Denver New York City Washington, DC
Bronze:	Albuquerque Baltimore Charlotte Columbus Fresno Indianapolis Kansas City Long Beach Louisville Mesa Milwaukee New Orleans Oakland Omaha Philadelphia Raleigh Sacramento San Antonio San Jose Tulsa

Source: LAB 2011

Benchmarking Efforts in the United States

Bicycle Friendly Community Awards

Although they don't use the term "benchmarking," the **League of American Bicyclists (LAB)** has created a system for scoring cities based on a measure of "bicycle-friendliness"—all of the ways a community promotes and accommodates bicycling. The **Bicycle Friendly Communities** program began in 1995 and is an awards program that recognizes municipalities that actively support bicycling. Cities interested in receiving a "Bicycle Friendly Community" designation submit an application to the League. The application is scored by a committee that consults with national and local bicyclists. Since its redesign and relaunch in 2003, 452 communities have applied for Bicycle Friendly Community designation and 181 have been awarded in that time. Currently 179 are designated.

LAB's Bicycle Friendly Community program includes Bronze, Silver, Gold, and Platinum levels awarded based on how communities score in five categories including engineering, education, encouragement, enforcement, and evaluation. This program has been extremely valuable to incite a spirit of competition among communities to be designated "Bicycle Friendly." The program also forces communities to complete an in-depth application, which gives them an opportunity to evaluate where they stand and causes them to gather data on bicycling in their community.

Walking Friendly Communities

In 2010 the **Pedestrian and Bicycle Information Cente**r launched the **Walking Friendly Communities (WFC)** program, modeled after the BFC program described above. WFC is a national recognition program developed to encourage U.S. communities to support safer walking environments. The WFC program recognizes places that are working to improve conditions for walking, including safety, mobility, access, and comfort. In April 2011, eleven communities received some level of WFC award. Seattle, Austin, and Charlotte were the only cities in this report to receive a WFC award in this first round.

Benchmarking State Policies

The **National Center for Bicycling and Walking (NCBW)** conducted a one-time study between December 2002 and February 2003 to evaluate state Departments of Transportation (DOTs) accommodating bicycles and pedestrians. **"The Benchmarking Project"** focused on data from questionnaires sent to the Bicycle and Pedestrian Coordinator of state DOTs. NCBW identified four benchmarks: presence of statewide long-range plan for bicycle/pedestrian elements, accommodating bicycles into all transport projects, accommodating pedestrians into all state highway projects, and other special programs.

NCBW assessed whether each state met national standards for these Benchmarks. Results were reported as "Yes" or "No" for each state meeting all or part of the benchmark, and summarized by each benchmark. They concluded that most state DOTs did not meet the benchmarks they identified for bicycle and pedestrian planning, accommodation (design), and special programs. All four of the benchmarks they identified are addressed in some way in Chapter 5 of this report. Although the Alliance's surveys did not frame questions in the same way, its review and discussion of complete streets policies, Safe Routes to School, and other bicycle and pedestrian policies address many of the same issues covered in NCBW's report.

Since the release of the 2007 Benchmarking Report, the **League of American Bicyclists** began a **Bicycle Friendly States** program that also compares all 50 states to each other on a number of indicators of "bicycle friendliness." The Bicycle Friendly States scoring system is based on 75 items that evaluate how committed the states are to bicycling. The six main areas the questionnaire covers are legislation, policies and programs, infrastructure, education and encouragement, evaluation and plan-

Links to Other Benchmarking Efforts

Abroad:

Copenhagen's Bicycle Account
http://cphbikeshare.com/files/Bicycle%20Account%202008.pdf
Europe: BYPAD—Bicycle Policy Audit
http://www.bypad.org/citymap.phtml?id=548&sprache=en
Europe: Velo Mondial
http://www.velomondial.net/page_display.asp?pid=14
Europe: Urban Transport Benchmarking Initiative
http://www.transportbenchmarks.eu/
London: The State of Walking
http://www.walk21.com/uploads/File/WC%20conference%20London%20100507.pdf
Netherlands: The Cycle Balance
http://www.fietsersbond.nl
Bicycling in Quebec
http://www.velo.qc.ca/en/Bicycling-in-Quebec
Toronto: Benchmarking Toronto's Bicycle Environment
http://www.torontocat.ca/main/node/454
UK: Cyclists Touring Club Benchmarking
http://www.ctc.org.uk/desktopdefault.aspx?tabid=3774
Walk 21: Measuring Walking
http://www.measuring-walking.org/

U.S.—National

Bicycle Friendly Communities Program
http://bit.ly/16G4lT
Bicycle and Pedestrian Documentation Project
http://bikepeddocumentation.org/
Bike Score
http://www.bikescore.com/
The College Sustainability Report Card
http://www.greenreportcard.org/
National Center for Bicycling and Walking
http://www.bikewalk.org/pdfs/ncbwpubthereyet0203.pdf
PBIC's Walkability and Bikeability Checklist
http://www.walkinginfo.org/library/details.cfm?id=12
http://www.bicyclinginfo.org/library/details.cfm?id=3
Walk Friendly Communities
http://www.walkfriendly.org/
Walk Score
http://www.walkscore.com/

U.S.—Local

New York's Bicycling Report Card
http://transalt.org/files/newsroom/magazine/2008/winter/06-08.pdf
San Francisco's Report Card on Bicycling
http://www.sfbike.org/?reportcard
Oregon's Bicycle Friendly Communities Report Card
http://www.bta4bikes.org/at*work/reportcard.php
Seattle's Report Card on Bicycling
http://www.cascade.org/advocacy/bicycle_report_card.cfm
Texas's Benchmarking Study
http://www.biketexas.org/en/infrastructure/benchmark-study

ning, and enforcement. The League released their fourth annual ranking of Bicycle Friendly States in 2011. The League hopes this will promote bicycling by listing which states recognize and support bicycling as an active form of transportation and recreation. States may also apply for awards under this program to receive further recognition for their bicycling efforts. Upon winning awards, states may also receive technical assistance, feedback, and training to further their bicycling plans.

Evaluating Walkability and Bikeability of Communities

The **Pedestrian and Bicycle Information Center's Walkability and Bikeability checklists** are another means of evaluating conditions for bicycling and walking. These checklists are community tools that allow individuals to subjectively score their communities. The document invites individuals to go for a walk or bicycle ride with survey in hand and to rate their experience on a scale of 1 to 5 while checking off potential problems. The document then goes through each question and offers potential solutions to common problems and also provides a list of resources at the end. This survey could be useful for community stakeholders wishing to gain insight into "bikeability" or "walkability." It could also be used by advocates in coordinated education efforts or to raise public perception of a problem area.

Looking at Universities

The **Bicycle Friendly University (BFU)** initiative was launched in 2010 by the **League of American Bicyclists** under their Bicycle Friendly America program. The BFU program recognizes institutions of higher education for promoting and providing a more bicycle friendly campus for students, staff, and visitors. The BFU program also provides information and technical assistance to create great campuses for cycling. Universities are invited to apply for BFU designation and are scored for their efforts in engineering, encouragement, education, enforcement, and evaluation/planning. In March 2011, twenty universities throughout the United States were honored with some level of Bicycle Friendly University award.

The **College Sustainability Report Card** is another effort to compare and evaluate campuses, but with a broader focus on all sustainability activities. The fifth report card, released in October 2010, scored more than 300 universities on administration, climate change and energy, food and recycling, green building, student involvement, transportation, endowment transparency, investment priorities, and shareholder engagement. The report relies on data from publicly available documentation, and from three surveys sent to school administrators. As of the most recent survey, 37% of schools earned an "A" grade in transportation. Key findings in the transportation category included:

- Bicycle-sharing programs have been instituted at 50 percent of schools.
- Car-sharing programs are available at 51 percent of schools.
- Reduced-fare passes for public transit are offered at 61 percent of schools.
- Hybrid or other alternative-energy vehicles are used in 86 percent of school fleets.

Walk Score Ranking of Cities

1. New York
2. San Francisco
3. Boston
4. Chicago
5. Philadelphia
6. Seattle
7. Washington, DC
8. Miami
9. Minneapolis
10. Oakland
11. Long Beach
12. Portland, OR
13. Los Angeles
14. Baltimore
15. Milwaukee
16. Denver
17. Cleveland
18. San Diego
19. San Jose
20. Atlanta
21. Omaha
22. Detroit
23. Houston
24. Sacramento
25. Las Vegas
26. Fresno
27. Tucson
28. Albuquerque
29. Columbus
30. Dallas
31. Austin
32. Tulsa
33. Phoenix
34. Colorado Springs
35. Mesa
36. Raleigh
37. Arlington
38. Wichita
39. Virginia Beach
40. San Antonio
41. Louisville
42. Memphis
43. Kansas City, MO
44. El Paso
45. Indianapolis
46. Nashville
47. Fort Worth
48. Oklahoma City
49. Charlotte
50. Jacksonville

Source:
http://www.walkscore.com/rankings
December 2011

- The average grade for the Transportation category was "B."

Although bicycling is a small component of this overall survey, there may be potential for future cooperation between the Benchmarking Project and this effort to collect more information and set benchmarks for how universities are promoting bicycling and walking.

National Bicycle and Pedestrian Documentation Project

Although not a benchmarking project per se, the **National Bicycle and Pedestrian Documentation Project (NBPD)** is addressing a critical component of all benchmarking efforts for bicycling and walking: trip counts. A more accurate and standardized way of measuring bicycling and walking trips would result in far more accurate benchmarking results. The National Bicycle and Pedestrian Documentation Project, coordinated by the Institute of Transportation Engineers, sets detailed standards and guidelines and provides tools for performing bicycle and pedestrian counts and surveys in communities. The objectives of the project are to:

(1) *Establish a consistent national bicycle and pedestrian count and survey methodology, building on the "best practices" from around the country, and publicize the availability of this free material for use by agencies and organizations online.*

(2) *Establish a national database of bicycle and pedestrian count information generated by these consistent methods and practices.*

(3) *Use the count and survey information to begin analysis on the correlations between various factors and bicycle and pedestrian activity. These factors may range from land use to demographics to type of new facility.*

As of January 2009, the project had collected 310 counts in about 93 different communities across the nation. NBPD has had a great variety of cities submit data. Large cities like San Jose, New York, Boston, and Portland have sent counts as well as smaller cities like San Rafael. Like the Alliance's Benchmarking Project, NBPD is working toward improving data collection and consistency in order to better understand influences and improve facilities and programs.

Scoring Walkability

Since the release of the 2007 Benchmarking Report, a new effort has launched to measure the walkability of cities. **Walk Score**, launched in July 2007, is a tool that "helps people find walkable places to live." Walk Score calculates the walkability of an address, or city, using a patent-pending system for measuring walkability. The calculator locates nearby stores, restaurants, schools, parks, and so on, to determine how close destinations are and determine how easy it is to get places by walking.

Since its launch, almost 6 million addresses have been served and Walk Score has been featured in over 500 newspaper articles and 50 TV segments. According to Walk Score, "Our vision is for every property listing to read: Bedrooms: 3 Baths: 2 Walk Score: 84. We want walkability and transportation costs to be a key part of choosing where to live." Walk Score has also used its methods to rank the 50 largest U.S. cities on walkability.

As of July 2011, the makers of Walk Score have been developing a version for biking called Bike Score. The public is invited to vote on what should be included in the Bike Score calculation at http://www.bikescore.com.

Local Efforts

Efforts to measure the state of bicycling locally have also been undertaken by local advocacy organizations. Alliance member organizations including Transportation Alternatives (New York City), San Francisco Bicycle Coalition, the Bicycle Transportation Alliance (Oregon), Cascade Bicycle Club, and BikeWalk Virginia have all created report cards for rating their communities at least once. Since the 2010 Benchmarking Report, new local benchmarking efforts have begun in Texas, Wisconsin, and Indiana.

The results of these efforts are that communities receive credit for areas where they are doing well, and areas needing improvement are identified. These efforts, many of which are modeled after this Benchmarking Report, serve as a benchmarking tool for cities to evaluate themselves and to use these data to measure progress over time.

Transportation Alternatives Bicycling Report Card

Transportation Alternatives (T.A.), the New York City bicycle, pedestrian, and transit-advocacy organization, has the longest running report card for bicycling among U.S. cities. In 2009, T.A. published their 12th annual **NYC Bicycling Report Card**, assigning three grades to eight “bicycle basics” including bicycling environment, safety, and parking, among others. T.A. assigns one grade based on government effort and one grade based on their assessment of the reality on the streets. A third grade is assigned by an Internet public opinion poll that received 1,200 responses for the last report. According to T.A., the purpose of the report card is “to provoke and encourage our politicians and government agencies to make NYC safer and more convenient for current bicyclists and more inviting for future ones.” This report card provides a useful and provoking annual assessment of bicycling conditions and progress being made toward a more bicycle friendly New York City.

San Francisco Bicycle Coalition Report Card on Bicycling

In 2006, the **San Francisco Bicycle Coalition (SFBC)**, San Francisco’s bicycle advocacy organization, published its first **Report Card on Bicycling**. Unlike T.A.’s Bicycling Report Card, SFBC relied completely on survey responses from bicyclists in San Francisco. The survey was answered by 1,151 individuals and addressed topics such as bicycling environment, safety, theft, and transit connections. The survey also collected information on topics such as frequency and types of bicycle trips and what prevents people from bicycling more than they do. The SFBC gave San Francisco a "B–" overall and included recommendations for the city to improve the score. According to the SFBC, the report card is “an instrument to hold (our) local decision makers accountable for their stated commitments to boosting bicycling rates and safety and making bicycling a mainstream transportation mode.” In 2008 the SFBC published their second biennial report card after surveying over 1,800 San Francisco bicyclists during April 2008. In addition to the survey results, this second report also included a variety of other measurements and statistics from local sources.

BTA’s Bicycle Friendly Communities Report Card

The **Bicycle Transportation Alliance (BTA)**, Oregon’s statewide bicycle advocacy organization, produced its first **Bicycle-Friendly Community Report Card** in 2002. Grades were given to 20 of Oregon’s largest communities based on such things as quality and quantity of bicycle facilities, encouragement of bicycling, established safety programs, and feedback from community bicycle riders. The 20 communities received a letter grade ranging from A– to D–. A discussion highlighted the good, the bad, and the opportunities to increase bicycling in various Oregon regions. According to the BTA, their report was “designed to help communities assess their commitment to bicycling as both recreation and transportation.”

Cascade Bicycle Club's Report Card on Bicycling

Seattle had their first ever **Report Card on Bicycling** published in 2009 by the **Cascade Bicycle Club**. The report card, largely modeled after SFBC's efforts, reports on both local bicycling data from government sources and on the results of a local survey of 600 Seattle residents. The report card grades Seattle on four categories: participation, network, support facilities, and safety. Each of these categories was further divided into subcategories derived from surveys and government data. Each subcategory grade determined the category grades, and the grades of the four categories were averaged to give Seattle an overall grade of "B." According to the Cascade

Bicycle Club, "ultimately, the findings identified in this Report Card will drive Cascade's future advocacy efforts to ensure that our cyclists' concerns are at the forefront of our agenda."

Benchmarking across the State of Virginia

In 2009, **BikeWalk Virginia**, in cooperation with the Virginia Department of Health, Virginia Department of Motor Vehicles (DMV), and Virginia Department of Transportation (VDOT), released the first-ever comprehensive report of bicycling and pedestrian planning, resources, accommodations, and safety in the state of Virginia. The report was funded by a DMV safety grant. BikeWalk Virginia surveyed 39 cities, 95 counties, and 157 incorporated towns in Virginia.

They developed a new measure, the Virginia Active Transportation Index (VATI), to provide a "comprehensive picture of biking and walking resources in each locality." Each locality was scored (from 0 to a perfect score of 11) on the index based on the number of resources they reported, which included: comprehensive transportation plan, bicycle plan, pedestrian plan, greenway plan, bicycle advisory committee, pedestrian advisory committee, greenway advisory committee, law requiring persons 14 and under to wear a helmet, paved bicycle trails, and striped bike lanes. Findings also include identification of localities that reported receiving an Enhancement Grant from the Virginia Department of Transportation. According to BikeWalk Virginia, "The report established a valid benchmark against which progress in expanding resources can be measured." The organization plans to conduct continuing surveys and update the report every two years.

Bike Texas Benchmarking

Bike Texas, Texas' statewide bicycling advocacy organization, initiated a statewide benchmarking project in 2010 with consultation from the Alliance. Their study, unpublished as of this report, will reveal baseline data from at least the 30 largest Texas cities. Measures include pedestrian and bicycle infrastructure, education and advocacy programs, funding, mode share, and safety.

Bicycle Benchmarking in Wisconsin

The Bicycle Federation of Wisconsin released their first benchmarking report on the state of bicycling in Wisconsin in 2011. The report, released at the 2011 Wisconsin Bike Summit, reveals data on bicycling ridership, facilities, education, and encouragement, and more based on data from local, state, and national sources. The Bicycle Federation of Wisconsin plans to release this report, based on the Alliance's Benchmarking Project, every other year.

Benchmarking Together

All efforts described above provide inspiration or direct knowledge to inform the Alliance's Benchmarking Project. The Alliance will continue to track other benchmarking efforts and encourage local communities to use the results of this report to support their own benchmarking efforts.

Appendix 7: Corrections to 2010 Benchmarking Report

The Alliance for Biking & Walking and our project team of advisors makes every effort to ensure the accuracy of data contained in this report. The self-reported nature of state and city data can lead to discrepancies from year to year, especially as respondents may change and interpret questions differently. In our effort to ensure accurate tracking and reporting of data, a number of responses to 2010 Benchmarking Report surveys have been updated. These corrections are reflected in the data analysis contained in this report. Below is a complete list of all corrections to the initial printed version of the 2010 report released in January 2010. Corrections are organized by chapter and page number.

Acknowledgments

- Teton Valley Trails and Pathways was misspelled

4: Policies and Provisions

Page 60:
- Published goal to increase walking—Response corrected to "no": Arkansas, New Hampshire, North Dakota, South Carolina
- Published goal to increase bicycling—Response corrected to "no": Arkansas, New Hampshire, South Carolina
- Published goal to decrease pedestrian fatalities—Response corrected to "no": New Hampshire
- Published goal to decrease bicycle fatalities—Response corrected to "no": New Hampshire, Wyoming
- Master Plan adopted for bicycling—Response corrected to "no": Iowa, North Dakota, Oklahoma, South Carolina
- Master Plan adopted for walking—Response corrected to "no": Hawaii, Maine, Oklahoma, South Carolina
- Bicycle and Pedestrian Advisory Committee—Response corrected to "no": New Hampshire

Page 61:
- Published goal to increase walking—Response corrected to "no": Colorado Springs, Dallas
- Published goal to decrease pedestrian fatalities—Response corrected to "no": Atlanta, Colorado Springs, Dallas, Miami
- Published goal to decrease bicycle fatalities—Response corrected to "no": Atlanta, Colorado Springs, Dallas
- Master Plan adopted for bicycling—Response corrected to "no": Memphis, Mesa, Tulsa; Response corrected to "yes": Charlotte
- Master Plan adopted for walking—Response corrected to "no": Charlotte, Memphis, Mesa, Phoenix, San Diego, Tulsa
- Bicycle and Pedestrian Advisory Committee—Response corrected to "no": Las Vegas, Memphis, Philadelphia

Pages 63-65:
- Complete Streets Policy—Response corrected to "no": El Paso, Kentucky

Pages 64-65:
- Complete Streets Policy—Corrected totals: Cities (13 with policies), States (17 with policies)

Page 64:

- Max number of car parking spaces for new buildings—Response corrected to “no”: Chicago, Memphis
- Bike parking requirements in buildings/garages—Response corrected to “no”: Las Vegas
- Bike parking requirements in new buildings—Response corrected to “no”: Las Vegas
- Bike parking at public events—Response corrected to “no”: Baltimore, Colorado Springs, Columbus, Milwaukee

Page 68:

- Provides SRTS funding beyond federal—Response corrected to “no”: Arizona, Massachusetts

Pages 88 and 90:

- Existing miles of on-street bike lanes per square mile (total miles)—Atlanta corrected to 0.1 mile per square mile (19 miles); Austin corrected to 0.6 mile per square mile (137 miles); Fort Worth corrected to 0.1 mile per square mile (13.8 miles); New York City corrected to 1.1 miles per square mile (341 miles); Phoenix corrected to 0.8 mile per square mile (365 miles); San Diego corrected to 1.0 mile per square mile (309.4 miles); Tulsa corrected to 0.1 mile per square mile (8.6 miles)
- Existing miles of multi-use paths per square mile (total miles)—Atlanta corrected to 0.2 mile per square mile (29 miles); Fort Worth corrected to 0.2 mile per square mile (57.3 miles); Louisville corrected to 0.4 mile per square mile (24.3 miles); Milwaukee corrected to 0.03 mile per square mile (3 miles); New York City corrected to 1.0 mile per square mile (295 miles); Virginia Beach corrected to 0.3 miles per square mile (74.7 miles)
- Existing miles of signed bicycle routes per square mile (total miles)—Fort Worth corrected to 0.1 mile per square mile (38.9 miles); Los Angeles corrected to “unknown” miles per square mile (“unknown” miles); Louisville corrected to 1.4 miles per square mile (89.8 miles); Phoenix corrected to 0.3 mile per square mile (122 miles); Tucson corrected to 0.5 mile per square mile (90 miles); Tulsa corrected to 0.5 mile per square mile (82.6 miles); Virginia Beach corrected to 0.3 mile per square mile (75 miles)

Pages 89-90:

- Existing total miles of bicycle facilities per square mile—Atlanta corrected to 0.6 mile per square mile; Austin corrected to 1.7 miles per square mile; Fort Worth corrected to 0.4 mile per square mile; Louisville corrected to 2.2 miles per square mile; Milwaukee corrected to 2.4 miles per square mile; New York City corrected to 2.1 miles per square mile; Phoenix corrected to 1.5 miles per square mile; San Diego corrected to 1.2 miles per square mile; Tucson corrected to 3.4 miles per square mile; Tulsa corrected to 1.1 miles per square mile; Virginia Beach corrected to 0.6 mile per square mile

Page 91:

- Shared lane markings—Response corrected to “no”: Dallas
- Woonerf/living streets—Response corrected to “no”: Dallas

5: Education and Encouragement

Page 97:

- Annual statewide bike/ped conference—Response corrected to “no”: Illinois

Page 100:

- State-sponsored ride to promote bicycling/activity—Response corrected to “no”: Maine

Page 106:
- Youth bike education courses—Response corrected to "no": Omaha
- Adult bike education courses—Response corrected to "no": Fort Worth, Las Vegas
- Bike to Work Day event—Response corrected to "no": Las Vegas
- Open Streets/Ciclovia event—Response corrected to "no": Baltimore, Chicago, Phoenix

Page 106:
- City-sponsored bike ride—Response corrected to "no": Las Vegas; Response corrected to "yes": Phoenix

Page 110:
- City-sponsored bike ride, number of participants—New York City response corrected to 30,000 participants for all three years (2006-2008); Phoenix ride added: 924 participants (2006), 912 participants (2007), 1,023 participants (2008)

6: Grassroots Advocacy

Page 120:
- Residents per one member—Colorado should be 693 adults per one member

Appendix 5: Resources

- The correct website for the National Bicycle and Pedestrian Documentation Project is http://bikepeddocumentation.org/

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